

# A Tutorial on Open-Source Analog Mixed-Signal Chip Design with the ihp-sg13g2 Open-PDK

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2026-08-24

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### ! Important

This tutorial and the underlying template repository require the [IIC-OSIC-TOOLS](#) container with tag 2026.08 or later. All commands below must be executed from within the container, and the working directory must be a folder containing the corresponding Makefile.

### i Note

We happily accept [pull requests](#) to fix typos or add content! If you want to discuss something that is not clear, please [open an issue](#)!

## 1 Tutorial Overview

This tutorial is a step-by-step walk-through of the open-source analog mixed-signal (AMS) chip flow based on the [ihp-sg13g2-ams-chip-template](#) repository for the ihp-sg13g2 130nm Open-PDK. It covers:

- [Overview](#) and [Installation](#) of [IIC-OSIC-TOOLS](#)
- [Digital macro design](#)
  - Design files written in SystemVerilog
  - Linting with [Verilator](#)
  - Simulation with [Icarus Verilog](#) and [cocotb](#)
  - Viewing waveforms with [GTKWave](#) and [Surfer](#)
  - Synthesis, placement, routing, and verification with [Yosys](#), [OpenROAD](#), and [Libre-Lane](#)

- [Analog macro design](#)
  - Schematic entry in [Xschem](#)
  - Simulation with [Ngspice](#) and [VACASK](#)
  - Monte Carlo / mismatch characterisation with [CACE](#)
  - Layout in [KLayout](#)
  - Verification with DRC, LVS, PEX using [KLayout](#), [Magic](#) + [Netgen](#)
- [Top-level assembly and padframe generation](#) with logos and fill structures using SystemVerilog and the [LibreLane](#) flow.

The example design used throughout the tutorial contains:

- Two digital `counter_top` macros (8-bit synchronous up-counters)
- Two `inverter_top` macros, one used as a 4-channel CMOS digital inverter, one as a 2-channel analog inverter
- One `RM_IHPSG13_1P_1024x32_c2_bm_bist` single-port SRAM macro
- A 32-pad padframe (8 pads per side) with I/O, power, analog, and bidirectional pads
- The top-level chip assembly with power distribution network (PDN), logos, and fill structures

The tutorial ends with a set of exercises that apply the same flow to a modified counter, a ring oscillator using the inverters, and a modified top-level floorplan and pinout. This way, all important steps of the flow are covered in a hands-on manner.



#### Tip

The tutorial can be used both for self-study and as the basis for a workshop. Each section can be followed independently.



#### Note

After the tutorial, readers should be able to use the template repository for custom silicon projects and understand how to apply the open-source tools in the IIC-OSIC-TOOLS container to design, simulate, build, and verify AMS chips with the ihp-sg13g2 Open-PDK.

## 1.1 Guidelines & Cheatsheets

The following guidelines and cheatsheets should be helpful for the tutorial, for the use of the template repository in general, and for implementing your own custom silicon projects.

- [Designer's Guidelines](#)
- [Linux Cheatsheet](#)
- [SystemVerilog Cheatsheet](#)



- [Verilog Cheatsheet](#)
- [Xschem Cheatsheet](#)
- [Ngspice Cheatsheet](#)
- [KLayout Cheatsheet](#)
- [KLayout Productivity Suite / Plugins](#)
- [LibreLane Cheatsheet](#)
- [ihp-sg13g2 Layout Calculation Cheatsheet](#)
- [ihp-sg13g2 LV NMOS / PMOS and HV NMOS / PMOS Techsweeps](#)

## 2 Tools

Numerous open-source chip design tools cover the entire development workflow for purely digital, purely analog, or mixed analog-digital chips. Managing these many tools regarding installation, updates, and maintenance is complex, especially given the rapid pace of change. Therefore, the [Institute for Integrated Circuits and Quantum Computing](#) at [Johannes Kepler University \(JKU\)](#) Linz, Austria, decided to build an all-in-one Docker container ([IIC-OSIC-TOOLS](#)) that is updated monthly and supports all essential tools and PDKs. This eliminates the need for the time-consuming individual installation of many tools, and a reproducible development environment (important for reproducing past tapeouts) is available within a GNU/Linux-based virtual machine.

### 2.1 IIC-OSIC-TOOLS Overview

Figure 1 shows a rough overview of the most important tools in the IIC-OSIC-TOOLS container. Additional tools are also included, which are documented on the associated [GitHub page](#). The open-source AMS flow used in this tutorial relies on several tools, namely [Xschem](#), [Magic](#), [Netgen](#), [KLayout](#), [Ngspice](#), [VACASK](#), [CACE](#), [LibreLane](#), [Yosys](#), [OpenROAD](#), [Verilator](#), [cocotb](#), [GTKWave](#), and [Surfer](#). The container also ships with four PDKs: [SkyWater sky130A](#), [GlobalFoundries gf180mcuD](#), [IHP ihp-sg13cmos5l](#) and [IHP ihp-sg13g2](#) used by this template.

### 2.2 IIC-OSIC-TOOLS Installation

The container runs on Linux, macOS, and Windows. The installation instructions are documented in the container's [README.md](#). For this tutorial, we recommend the quick one-liner installation.

Additional video walk-throughs are available for Linux and Windows.

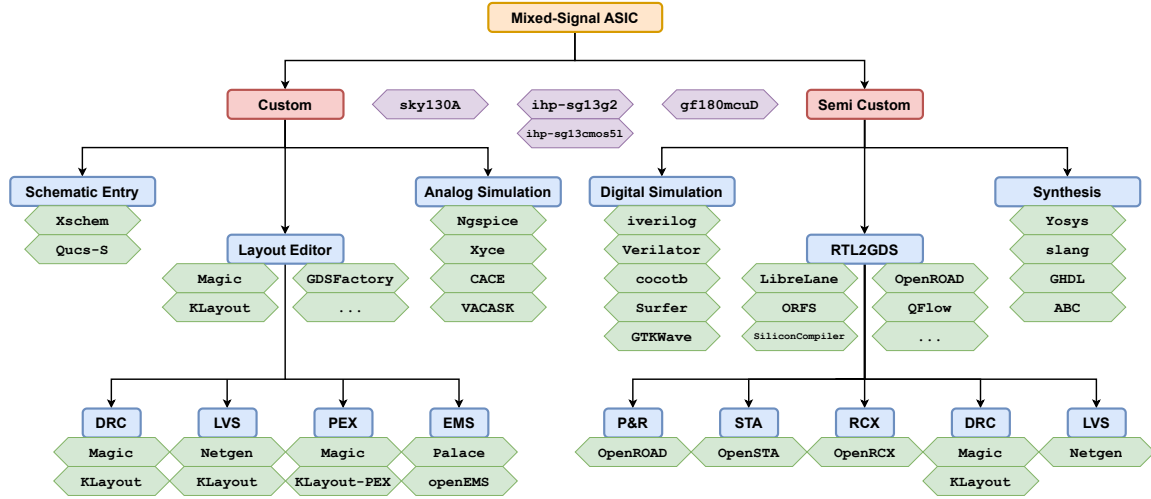


Figure 1: Overview of the most important open-source tools and PDKs shipped with the IIC-OSIC-TOOLS container.

### **i** Ubuntu / Linux installation

- [IIC-OSIC-TOOLS installation on Ubuntu](#)

### **i** Windows installation

- [IIC-OSIC-TOOLS installation on Windows \(from 37:20 to 45:15\)](#)

## 2.3 PDK Selection

As mentioned above, the container includes four Open-PDKs. To activate or switch the active PDK, the following two options are available. More detailed instructions can be found in the container's [README.md](#).

1. **.designninit file:** Copy the `.designninit` file referencing the `ihp-sg13g2` PDK into your `designs/` directory and restart the container. The PDK is then picked automatically every time the container is launched in that directory.

2. **On-the-fly switch:** Run

```
sak-pdk ihp-sg13g2
```

inside the container. This switches the active PDK for the current session without modifying any file. When called without arguments (`sak-pdk`), a list of installed PDKs

is shown.

#### Tip

By default, the container is already configured for `ihp-sg13g2`, so both options above are only needed when switching between PDKs. However, since this tutorial can also be used for all other PDKs, it is worth knowing how to switch the active PDK.

## 3 Repository Overview

### 3.1 Purpose

This Makefile-driven repository simulates, builds, and fully verifies (DRC, LVS, PEX) a complete analog mixed-signal chip for the `ihp-sg13g2` 130nm Open-PDK, including padframe generation and top-level assembly. It uses:

- [LibreLane](#) for digital macro hardening, padframe generation and top-level assembly
- [Xschem](#) for schematic entry
- [Ngspice](#), [VACASK](#) and [CACE](#) for analog simulation
- [KLayout](#) for viewing and routing of the layout
- [Magic](#) + [Netgen](#) and [KLayout](#) for DRC, LVS and PEX verification
- [SystemVerilog](#), [cocotb](#), [GTKWave](#) and [Surfer](#) for digital simulation

The repository is the starting point for your own custom silicon and provides a universal design flow solution: Just clone the repo, enter the IIC-OSIC-TOOLS container, and run `make all` to get a tapeout-ready analog mixed-signal chip. Focus on your design and do not worry about the tools and the design flow!

Furthermore, it serves as a regression test for the above-mentioned open-source tools and their dependencies using the `ihp-sg13g2` Open-PDK.

### 3.2 Examples

Examples based on this template are:

- [TinyWhisper](#): An Open-Source Fully-Integrated Multi-Mode Short-Wave Transmitter for Amateur Radio Applications in 130-nm CMOS
- [SPARX](#): An Open-Source, Automated, Programmatically Generated, Frequency-Scalable Six-Port Receiver in 130-nm CMOS
- [Multi-Project Chip](#) for the wafer.space gf180mcuD MPW run

### 3.3 Directory Structure

The repository is organised into the following top-level folders. The full listing can be found in the [top-level README.md](#).

| Folder                        | Purpose                                                                                                                                                                        |
|-------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <a href="#">doc/</a>          | Designer documentation, including <a href="#">specifications</a> , <a href="#">pinout</a> , <a href="#">floorplan</a> , and the <a href="#">PDK and tool cheatsheets</a> .     |
| <a href="#">flow/</a>         | LibreLane top-level configuration with <a href="#">config.yaml</a> , <a href="#">pdn_cfg.tcl</a> , <a href="#">chip_top.sdc</a> , plus the <a href="#">ArtistIC</a> logo flow. |
| <a href="#">ip/</a>           | External IP cells including bondpads, custom IO cells, and logos.                                                                                                              |
| <a href="#">layout/</a>       | Compressed GDS streams of the top-level chip, including <a href="#">chip_top.gds.gz</a> and <a href="#">chip_top_logo_fill.gds.gz</a> .                                        |
| <a href="#">LICENSES/</a>     | License texts referenced by the SPDX headers and <a href="#">REUSE.toml</a> annotations.                                                                                       |
| <a href="#">macros/</a>       | Recursive-style macros such as <a href="#">counter/</a> and <a href="#">inverter/</a> , each with its own Makefile and README.                                                 |
| <a href="#">netlist/</a>      | Synthesis, schematic, layout, and PEX netlists used for top-level LVS and simulation.                                                                                          |
| <a href="#">packaging/</a>    | Automated bondplan generation: config, flow script, package and bondplan GDS, bonding diagrams, and bond report (see <a href="#">Section 7.9</a> ).                            |
| <a href="#">release/</a>      | Tapeout submission artifacts grouped by version.                                                                                                                               |
| <a href="#">render/</a>       | Chip and macro renders used in the documentation.                                                                                                                              |
| <a href="#">rtl/</a>          | Top-level RTL sources, including <a href="#">chip_top.sv</a> and <a href="#">chip_core.sv</a> .                                                                                |
| <a href="#">schematic/</a>    | Xschem schematics and symbols for the top-level chip.                                                                                                                          |
| <a href="#">scripts/</a>      | Helper Python and shell scripts for rendering, logo placement, and plotting.                                                                                                   |
| <a href="#">testbenches/</a>  | Xschem and cocotb testbenches for the top-level chip.                                                                                                                          |
| <a href="#">tutorial/</a>     | This Quarto tutorial and its supporting materials.                                                                                                                             |
| <a href="#">verification/</a> | DRC, LVS, and signoff reports.                                                                                                                                                 |

### 3.4 Workflow

The flow is driven by Makefiles. After cloning the repository and entering the IIC-OSIC-TOOLS container, every step is invoked via a **make** target. The default target of every **Makefile** is **help**, so **make** (without arguments) prints the list of available targets. After entering the container, the starting folder is **/foss/designs/**.

```
git clone https://github.com/iic-jku/ihp-sg13g2-ams-chip-template.git
cd ihp-sg13g2-ams-chip-template
make help                # show available targets at the top level
```

### 💡 Tip

Instead of `git clone`, you can start by clicking the green “Use this template” button on the GitHub page to create your own copy of the repository. This way, you can push your changes to your own GitHub repository and even make it public if you want to share your custom silicon design with the community.

Each macro under `macros/` has its own `Makefile` and `README.md`, including macro-specific targets. These targets are invoked by the top-level build flow, but you can also run them directly from inside each macro folder. Figure 2 shows how the `Makefile` targets across the repository are connected.

The colour scheme groups targets by the final deliverable they produce, so beginners can follow one colour from `make all` to the generated outputs:

- **Orange (top chip / main):** This branch includes `make all` and all targets that contribute to the top-level chip GDS. In the `build-top` chain of the `top-level Makefile`, these are `librelane-nodrc`, `copy-*`, `add-logo-fill`, and `render-gds`.
- **Yellow (bondpad IP):** This branch includes `build-bondpad` and all targets inside `ip/sg13g2_ip__bondpad_70x70/`.
- **Purple (logos IP):** This branch includes `build-logos` and the two sub-`Makefiles` under `ip/sg13g2_ip__jku/` and `ip/sg13g2_ip__jku_names/`.
- **Blue (digital macro):** This branch includes `build-counter` and the `macros/counter/` `Makefile`.
- **Green (analog macro):** This branch includes `build-inverter` and the `macros/inverter/` `Makefile`.
- **Red (packaging):** This branch includes `bondplan` and the automated bondplan flow in `packaging/` (see Section 7.9).
- **Grey (simulation and verification):** This branch includes intermediate targets that feed the coloured deliverables (`sim-all`, `build-all`, `build-macros`, `init-submodules`, `sim-*-cocotb`).

Solid arrows represent direct `$(MAKE) <target>` calls within a single `Makefile`. Dashed arrows represent calls that descend into a subdirectory: recursive `$(MAKE) -C <dir> all` calls into a sub-`Makefile` (one per coloured deliverable), or the Python bondplan flow in `packaging/`. The numbers on the second level indicate the execution order of `make all` (build, verify, simulate, package), and the vertical order inside the coloured boxes reflects the execution order of each sub-`Makefile`’s `all` target.

### 3.4.1 Top-level targets

The `top-level Makefile` provides:

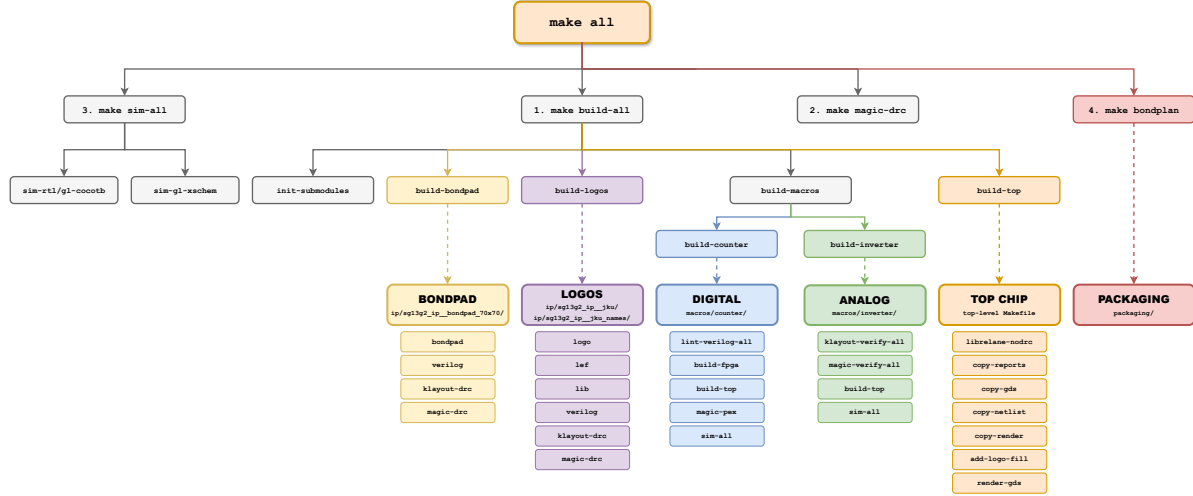


Figure 2: Overview of the Makefile targets that `make all` resolves to. Every coloured branch corresponds to one deliverable (top chip, bondpad, logos, digital macro, analog macro, packaging). The grey targets connect `make all` to those branches.

| Target                                                                         | What it does                                                                                                                                                                                      |
|--------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <code>make help</code>                                                         | Print the help banner and list every target with its description.                                                                                                                                 |
| <code>make open</code>                                                         | Open the design files of this folder in the <code>sak-open.py</code> file browser (needs the VNC/X11 desktop).                                                                                    |
| <code>make init-submodules</code>                                              | Initialise and update git submodules (for example the ArtistIC logo flow).                                                                                                                        |
| <code>make sim-rtl-cocotb /<br/>sim-gl-cocotb</code>                           | Run the RTL / gate-level (GL) cocotb testbench for the top-level.                                                                                                                                 |
| <code>make sim-gl-xschem</code>                                                | Run the gate-level Xschem testbench in batch mode (TB defaults to <code>&lt;CELL&gt;_tb_tran</code> ). Part of <code>sim-all</code> , but it may take a long time depending on the hardware used. |
| <code>make sim-view-cocotb</code>                                              | Open the latest cocotb waveform in GTKWave (or Surfer with <code>WAVEFORM_VIEWER=surfer</code> ).                                                                                                 |
| <code>make librelane /<br/>librelane-nodrc</code>                              | Run the LibreLane flow for the top-level (with / without DRC steps).                                                                                                                              |
| <code>make copy-reports /<br/>copy-gds / copy-netlist<br/>/ copy-render</code> | Copy the latest LibreLane artifacts back into the source tree.                                                                                                                                    |
| <code>make build-bondpad /<br/>build-logos</code>                              | Build the bondpad and the logos under <code>ip/</code> .                                                                                                                                          |
| <code>make build-counter /<br/>build-inverter /<br/>build-macros</code>        | Build the digital and analog macros individually or together.                                                                                                                                     |

| Target                                              | What it does                                                                                                      |
|-----------------------------------------------------|-------------------------------------------------------------------------------------------------------------------|
| <code>make build-top</code>                         | Run LibreLane on the top-level, copy back all artifacts, add logos and fill structures, and render the final GDS. |
| <code>make build-all</code>                         | Init submodules, build bondpad, build logos, build macros, and build top-level.                                   |
| <code>make klayout-verify /<br/>magic-verify</code> | Run DRC, LVS, and PEX for a given cell with KLayout or Magic.                                                     |
| <code>make bondplan</code>                          | Generate the bondplan (die in package, bondwires, pin table) in <code>packaging/</code> (see Section 7.9).        |
| <code>make all</code>                               | Full build, DRC verification, simulation, and bondplan generation.                                                |
| <code>make release<br/>VERSION=2.1.0</code>         | Copy GDS, netlists, and renders into <code>release/v.&lt;VERSION&gt;/</code> for tapeout submission.              |
| <code>make clean / make<br/>clean-all</code>        | Delete the generated files of the chip top-level, or of the chip top-level plus the IPs and the macros.           |

See the [top-level README](#) for more details and how to use the Makefile targets.

### 3.4.2 Digital macro targets

The [counter Makefile](#) provides:

| Target                                                       | What it does                                                                                                                                                                       |
|--------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <code>make open</code>                                       | Open the design files of this folder in the <code>sak-open.py</code> file browser.                                                                                                 |
| <code>make lint-verilog / make<br/>lint-verilog-all</code>   | Lint a single cell or the full counter design with Verilator.                                                                                                                      |
| <code>make sim-rtl-verilog</code>                            | Run the testbench with Icarus Verilog ( <code>&lt;CELL&gt;_tb.sv</code> , falling back to <code>&lt;CELL&gt;_tb.v</code> ).                                                        |
| <code>make sim-rtl-cocotb /<br/>sim-gl-cocotb</code>         | Run the cocotb RTL / gate-level simulation.                                                                                                                                        |
| <code>make sim-view-verilog /<br/>sim-view-cocotb</code>     | Open the latest waveform in GTKWave or Surfer.                                                                                                                                     |
| <code>make librelane (and -nodrc,<br/>-magicdrc, ...)</code> | Run the LibreLane flow for the counter macro.                                                                                                                                      |
| <code>make symbol-gl (FORCE=1)</code>                        | Scaffold the Xschem symbol <code>&lt;CELL&gt;.sym</code> from the ports of <code>netlist/pnl/&lt;CELL&gt;.pnl.v</code> . For a new macro, refuses to overwrite an existing symbol. |
| <code>make symbol-check</code>                               | Check that <code>&lt;CELL&gt;.sym</code> still describes those ports. Runs automatically before every XSPICE conversion.                                                           |

| Target                                                                       | What it does                                                                                                                                                                                     |
|------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <code>make generate-xspice</code>                                            | Convert the LibreLane-extracted SPICE netlist into an XSPICE model for Xschem.                                                                                                                   |
| <code>make sim-gl-xschem /<br/>sim-view-xschem</code>                        | Run the mixed-signal XSPICE simulation in Xschem (batch mode) and plot the results. TB defaults to <code>&lt;CELL&gt;_tb_tran</code> and <code>SCRIPT</code> to <code>plot_&lt;CELL&gt;</code> . |
| <code>make build-fpga</code>                                                 | Run the full FPGA flow: lint, synthesis, P&R, and bitstream generation. The board is selected with <code>BOARD=</code> and defaults to the pico-ice.                                             |
| <code>make build-top</code>                                                  | Run LibreLane, copy reports, netlists, and final views, generate the XSPICE model, and render the final GDS.                                                                                     |
| <code>make symbol-pex</code>                                                 | Build the Xschem PEX symbol <code>&lt;CELL&gt;_pex.sym</code> from <code>&lt;CELL&gt;.sym</code> . Runs automatically before every extraction.                                                   |
| <code>make klayout-pex /<br/>magic-pex<br/>(EXT_MODE=&lt;1\ 2\ 3&gt;)</code> | Extract the parasitics of the hardened macro from <code>final/gds/</code> into <code>netlist/pex/</code> .                                                                                       |
| <code>make all</code>                                                        | Run lint, the FPGA build, the full macro build flow, the parasitic extraction, and all simulations.                                                                                              |
| <code>make clean</code>                                                      | Delete all generated files and folders of the macro, including the FPGA outputs.                                                                                                                 |

See the [counter README](#) for more details and how to use the Makefile targets.

### 3.4.3 Analog macro targets

The [inverter Makefile](#) provides:

| Target                                                                                         | What it does                                                                                                                                                                                                                                               |
|------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <code>make open</code>                                                                         | Open the design files of this folder in the <code>sak-open.py</code> file browser.                                                                                                                                                                         |
| <code>make sim-xschem / make<br/>sim-view-xschem</code>                                        | Run an Xschem testbench (batch mode) and plot its results with Python. TB defaults to <code>&lt;CELL&gt;_tb_tran</code> and <code>SCRIPT</code> to <code>plot_&lt;CELL&gt;</code> , override with <code>TB=&lt;tb&gt; /<br/>SCRIPT=&lt;script&gt;</code> . |
| <code>make sim-cace</code>                                                                     | Run <a href="#">CACE</a> Monte Carlo / mismatch characterisations.                                                                                                                                                                                         |
| <code>make klayout-drc /<br/>magic-drc<br/>(DRC_LEVEL=&lt;precheck\ macro\ regular&gt;)</code> | Run DRC using KLayout or Magic.                                                                                                                                                                                                                            |
| <code>make klayout-lvs /<br/>magic-lvs</code>                                                  | Run LVS using KLayout or Magic + Netgen.                                                                                                                                                                                                                   |



| Target                                                                       | What it does                                                                                                                                   |
|------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------|
| <code>make symbol-pex</code>                                                 | Build the Xschem PEX symbol <code>&lt;CELL&gt;_pex.sym</code> from <code>&lt;CELL&gt;.sym</code> . Runs automatically before every extraction. |
| <code>make klayout-pex /<br/>magic-pex<br/>(EXT_MODE=&lt;1\ 2\ 3&gt;)</code> | Run parasitic extraction in C, CC, or full-RC mode.                                                                                            |
| <code>make klayout-verify /<br/>magic-verify</code>                          | Run DRC + LVS + PEX for a given cell.                                                                                                          |
| <code>make lef / make lib / make<br/>verilog</code>                          | Generate the LEF, the Liberty stub, and the Verilog stub for top-level integration.                                                            |
| <code>make build-top</code>                                                  | Build the analog macro deliverables: LEF, LIB, Verilog stub, GDS and rendered layout image.                                                    |
| <code>make all</code>                                                        | Run KLayout / Magic verification, the analog macro build flow, and all simulations.                                                            |
| <code>make clean</code>                                                      | Delete all generated files and folders of the macro, including the CACE outputs.                                                               |

See the [inverter README](#) for more details and how to use the Makefile targets.

## 4 Digital Macro Implementation

The [counter](#) macro is the digital reference macro of the template. It is a parameterisable synchronous up-counter with synchronous active-high reset implemented with SystemVerilog in [counter.sv](#). The top-level wrapper [counter\\_top.sv](#) converts the chip's active-low reset to an active-high signal. The default width is 8 bits, so the counter wraps from 255 back to 0.

### 4.1 Directory Structure

The counter design files are located in [macros/counter/](#), and the relevant folders are:

| Folder                       | Content                                                                                                                                           |
|------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------|
| <a href="#">rtl/</a>         | SystemVerilog sources: <a href="#">constants.sv</a> , <a href="#">counter.sv</a> , <a href="#">counter_top.sv</a> .                               |
| <a href="#">testbenches/</a> | <a href="#">verilog/</a> , <a href="#">cocotb/</a> , and <a href="#">xschem/</a> testbenches with pre-configured GTKWave and Surfer view files.   |
| <a href="#">flow/</a>        | LibreLane configuration ( <a href="#">config.yaml</a> , <a href="#">pin_order.cfg</a> , <a href="#">impl.sdc</a> , <a href="#">signoff.sdc</a> ). |

| Folder                                                    | Content                                                                                                                                                               |
|-----------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <a href="#">final/</a>                                    | Final deliverables (GDS, LEF, LIB per corner, NL, PNL, SPEF, VH) copied from the latest LibreLane run for top-level integration.                                      |
| <a href="#">fpga/</a>                                     | Optional FPGA emulation flow for three boards (pico-ice, iCEBreaker, ULX3S).                                                                                          |
| <a href="#">netlist/</a>                                  | <a href="#">nl/</a> , <a href="#">pnl/</a> , <a href="#">spice/</a> and <a href="#">xspice/</a> netlists (see <a href="#">LibreLane cheatsheet</a> for more details). |
| <a href="#">schematic/</a>                                | Xschem symbol of the macro for use in mixed-signal Xschem testbenches.                                                                                                |
| <a href="#">scripts/</a><br><a href="#">verification/</a> | Helper scripts (XSPICE conversion, plotting, layout rendering). Yosys, antenna, STA, IR-drop, DRC, LVS, and manufacturability reports.                                |
| <a href="#">render/</a>                                   | Macro renders for documentation.                                                                                                                                      |



Tip

Read the [counter README](#) first to get familiar with the folder layout and the available Makefile targets. Most of the commands below are documented there as well.

## 4.2 Reading the RTL

The macro is split into a parameterised counter and a top-level wrapper that converts the chip's active-low reset to the internal active-high reset of the counter.

### 4.2.1 `counter.sv`

The logic is implemented in [counter.sv](#). It is an  $N$ -bit synchronous up-counter with reset and enable, and it wraps at `CTR_MAX`.  $N$  defaults to 8, so the counter counts from 0 to 255 and then wraps back to 0.

### 4.2.2 `counter_top.sv`

[counter\\_top.sv](#) wraps the counter and converts the active-low reset (`reset_n_i`) to an active-high signal `reset`.

Shared defaults are defined in [constants.sv](#). For example, for the final implementation, the maximum counter value is defined in `constants.sv` with the variable `COUNTER_MAX_DEFAULT`. For the testbenches, `CLK_FREQ_DEFAULT` is defined there to set the default clock frequency

(e.g. 50 MHz) for the simulations. They are exposed as ``define` macros instead of a SystemVerilog package, because Yosys 0.64 cannot parse `import pkg::*` in a module header.

#### **i** Note

Open the three RTL files (using `gedit` or `vim`), follow and understand the counter through reset, hold, increment, and wrap-around.

## 4.3 Linting

Linting is done with `Verilator` and can be executed with the following commands.

```
cd macros/counter
make lint-verilog          # lint the full counter_top design
make lint-verilog CELL=counter # lint only the standalone counter
make lint-verilog-all      # lint counter and counter_top in sequence
```

`make lint-verilog-all` is also the lint step called by `make all`.

## 4.4 Simulation

### 4.4.1 RTL with SystemVerilog testbench (Icarus Verilog)

A SystemVerilog testbench (`counter_top_tb.sv`) checks reset, hold, increment, and wrap-around. Run it with the following commands.

```
make sim-rtl-verilog
make sim-view-verilog          # open in GTKWave (default)
make sim-view-verilog WAVEFORM_VIEWER=surfer
```

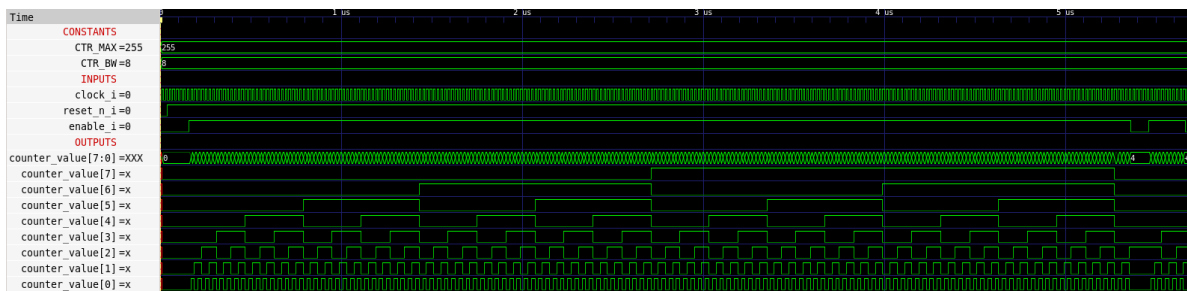


Figure 3: RTL Verilog waveform of `counter_top` in GTKWave.

#### 4.4.2 RTL and gate-level with cocotb

The `cocotb` testbench (`counter_top_tb.py`) checks the same four cases (reset, hold, increment, and wrap-around) as the SystemVerilog testbench, but implements them in Python.

```
make sim-rtl-cocotb          # RTL simulation
make sim-gl-cocotb          # post-synthesis gate-level simulation
make sim-view-cocotb        # view in GTKWave
make sim-view-cocotb WAVEFORM_VIEWER=surfer
```

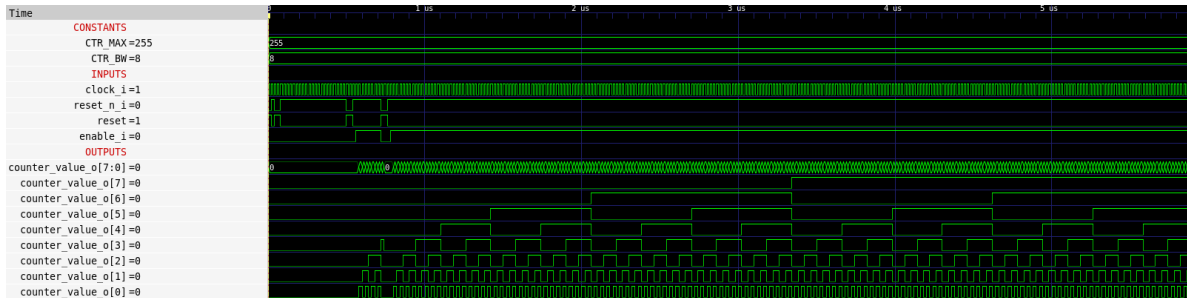


Figure 4: Gate-level cocotb waveform of `counter_top` in GTKWave.

The gate-level cocotb run uses the gate-level Verilog netlist produced after synthesis: `netlist/nl/counter_top.nl.v`. In this tutorial, this netlist for the counter already exists. For new designs, run `make build-top` first to execute the LibreLane flow. It also copies the netlists from the latest LibreLane run back into the source tree with `make copy-netlist`.

#### 4.5 LibreLane Hardening

The LibreLane flow synthesises, places, and routes the macro, runs the verification steps, and writes its final output files to `flow/final/`. The wrapper target `make build-top` runs the flow, renders the final GDS, and copies the reports, final folders, netlists, and render output back into the source tree.

```
make build-top
```

##### 💡 Tip

The macro can also be built as an FPGA bitstream (`fpga/`) with `make build-fpga`. `counter_top` is synthesised straight onto board pins, without a board-top wrapper, and three boards are supported: pico-ice and iCEBreaker (Lattice iCE40) and ULX3S (Lattice ECP5). Pick one with `make build-fpga` for the default pico-ice, or `make -C`

fpga BOARD=ulx3s all for another. IIC-OSIC-TOOLS carries the build chain for all three boards, so the bitstreams build inside the container, and only flashing needs the host, since the container has no USB access. Details are in [fpga/README.md](#). The FPGA flow is not used in this tutorial and is only listed here for reference.

## 4.6 Viewing the Macro

After the flow finishes, the macro layout can be inspected with KLayout. The container provides the `ke` alias for `klayout -e` (edit mode). With the following command, the final GDS from the latest LibreLane run is opened in KLayout. With the keys 0, 1, 2, ..., 9 you can view different layers as explained in the [KLayout cheatsheet](#) or in the [klayout-layer-shortcuts](#) plugin.

```
ke final/gds/counter_top.gds
```

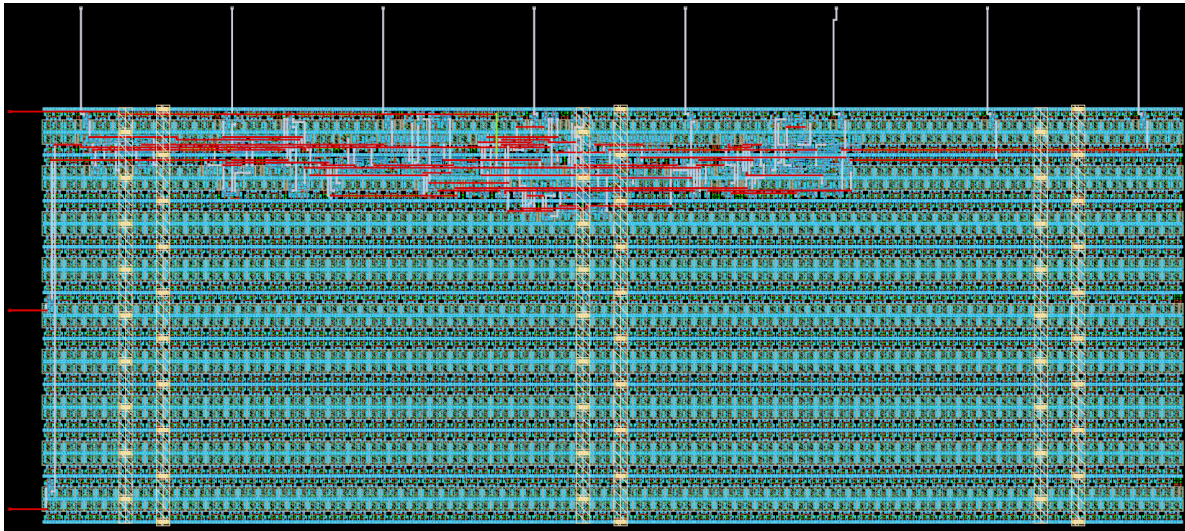


Figure 5: counter\_top layout in KLayout (200  $\mu\text{m}$   $\times$  100  $\mu\text{m}$ ).

The LibreLane render is copied to [render/img/counter\\_top\\_librelane.png](#), and the high-resolution renders from `sak-render.py` are saved as `counter_top_white.png` and `counter_top_black.png` in the same folder. The verification reports are written to [verification/](#).

## 4.7 RTL Mixed-Signal Simulation with Xschem

For an introduction to Ngspice and Verilog RTL co-simulation with Xschem, see Stefan Schippers' video: [Ngspice + Verilog Co-Simulation in Xschem](#). This RTL mixed-signal simulation is

currently not included in this repository, but it will be added in a future update.

## 4.8 Gate-Level Mixed-Signal Simulation with Xschem

A more accurate validation step is the gate-level mixed-signal simulation in Xschem with the Ngspice XSPICE event-driven engine. The LibreLane-extracted SPICE netlist (`netlist/spice/<CELL>.spice`, copied from the latest run by `copy-netlist`) is converted into an XSPICE model with the following command, which is also executed automatically as part of `make build-top`.

```
make generate-xspice
```

The conversion pipeline is `.spice`  $\rightarrow$  `.xspice` (`spi2xspice.py`), followed by a pin-reorder pass with `sak-pin-reorder.py` (a SAK tool installed in the IIC-OSIC-TOOLS container) to match the Xschem symbol (`schematic/xschem/counter_top.sym`).

### Note

The pins are matched **by name**: every pin in the Xschem symbol carries a `sim_pinname=<netlist_name>` property, so the alphabetically sorted ports of the extracted netlist are mapped correctly regardless of their order. Since `make build-top` regenerates the XSPICE model right after copying the netlists, the model always matches the current LibreLane run.

### 4.8.1 The symbol is a source file

The macro is hardened, so it has no schematic, and Xschem’s “make symbol from a schematic” (key `a`) has nothing to work from. `schematic/xschem/counter_top.sym` is drawn by hand, and everything downstream bends to it: the pin order of the XSPICE model and of the extracted PEX netlists is sorted to match it. It is therefore never regenerated by a build. The testbench wires to pin **coordinates**, and a pin one grid step off its wire floats without any error, so a build that moved pins would quietly break the simulation instead of failing it.

Two targets support that arrangement without taking ownership of the file away from you.

```
make symbol-gl      # scaffold schematic/xschem/<CELL>.sym, for a new macro
make symbol-check   # verify the committed symbol, runs inside generate-xspice
```

`symbol-gl` reads the ports of the powered netlist `netlist/pnl/<CELL>.pnl.v` and writes a first-cut symbol in which every port is a pin with its direction, its bus range and its `sim_pinname`. The geometry follows Xschem’s own symbol generator, so loading the result and saving it

again returns it unchanged. It is a scaffold: rename the pins to house style (`clock_i` becomes `di_clock`, `counter_value_o[0..7]` becomes `do_b[0..7]`), arrange them, draw the body, then commit it. Renaming is safe because `sim_pinname` holds the binding to the netlist. The target refuses to overwrite an existing symbol, because that is where the hand work lives.

`symbol-check` is the half that runs in the build. It compares the committed symbol against the same netlist and fails if a port was added, removed, renamed, given a different width, or drawn with the wrong direction. Its most useful check is that **every pin declares `sim_pinname`**: a single pin missing the property drops `sak-pin-reorder.py` out of name matching and into matching by position, which is wrong for an alphabetically sorted netlist and can mis-wire the macro without any error at all.

#### Tip

For a new digital macro the order is: harden it once, `make copy-netlist`, `make symbol-gl`, arrange the symbol in Xschem, and only then wire the testbench to it and run `make all`.

### 4.8.2 Running the gate-level testbench

The Xschem testbench (`counter_top_tb_tran.sch`) can be opened and simulated interactively.

```
cd testbenches/xschem
xschem counter_top_tb_tran.sch
```

In Xschem, hold **Ctrl** and click the **Simulate** button in the schematic to launch the simulator. Once the simulation has finished, you can hold **Ctrl** and click the **Load waves** arrow to load the simulation results into the integrated waveform viewer.

#### Tip

This testbench also shows multiple ways to connect and display bus signals in Xschem. For more information, refer to the [Xschem documentation: Use Bus/Vector notation for signal bundles / arrays of instances](#).

#### Note

The green arrows in Xschem are called “launchers” and are essentially a series of Tcl commands. Since Xschem is fully scriptable via Tcl, repetitive tasks can be easily automated. Select a launcher and press **q** (Properties) to inspect or modify its underlying Tcl commands. This is also the beauty of Xschem and the reason why it is possible to



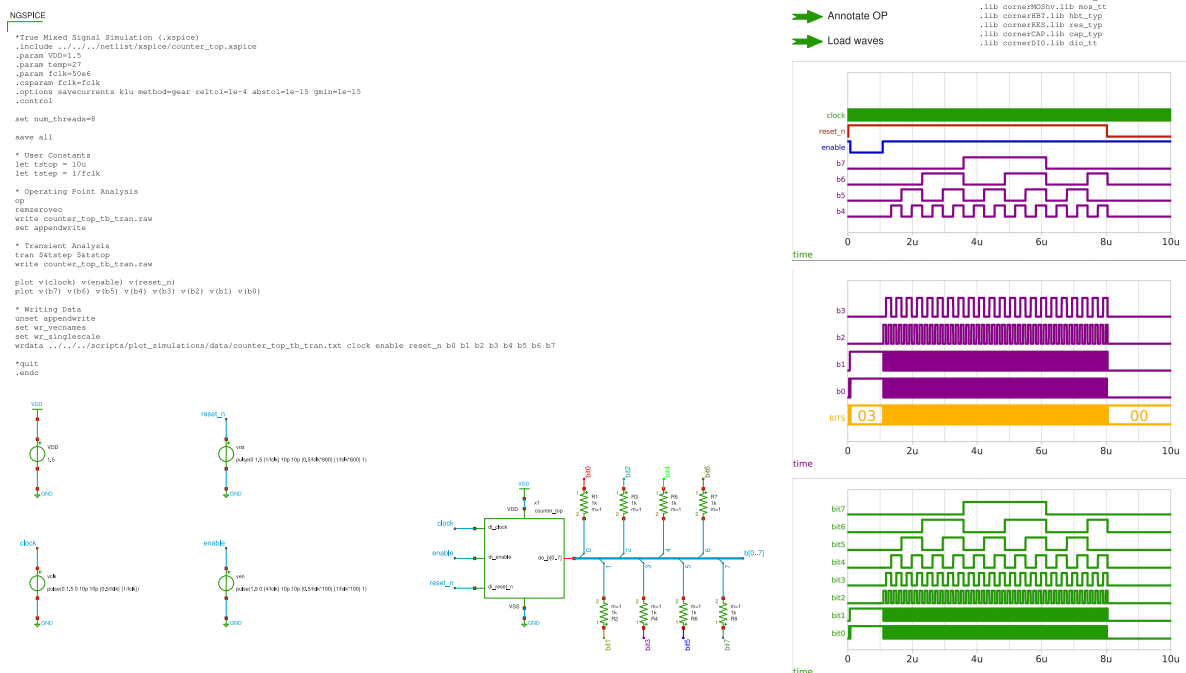


Figure 6: Xschem mixed-signal gate-level testbench counter\_top\_tb\_tran.

also run simulations non-interactively, generate netlists, and run LVS with only Makefile targets, as you will see below.

The same simulation can also be run non-interactively from the Makefile, and the result can be plotted from Python.

```

make sim-gl-xschem
make sim-view-xschem SCRIPT=plot_counter_top

```

sim-gl-xschem runs in batch mode: it netlists the testbench with xschem netlist and then calls ngspice -b directly, so make blocks until the run finishes and sees its exit status. The testbench is selected with TB=<testbench\_name> (default: <CELL>\_tb\_tran). Since the run is headless, the plot commands in the testbench's .control block are a no-op. The testbench exports its results with wrdata to testbenches/xschem/plot\_simulations/data/, from where sim-view-xschem plots them with Matplotlib.



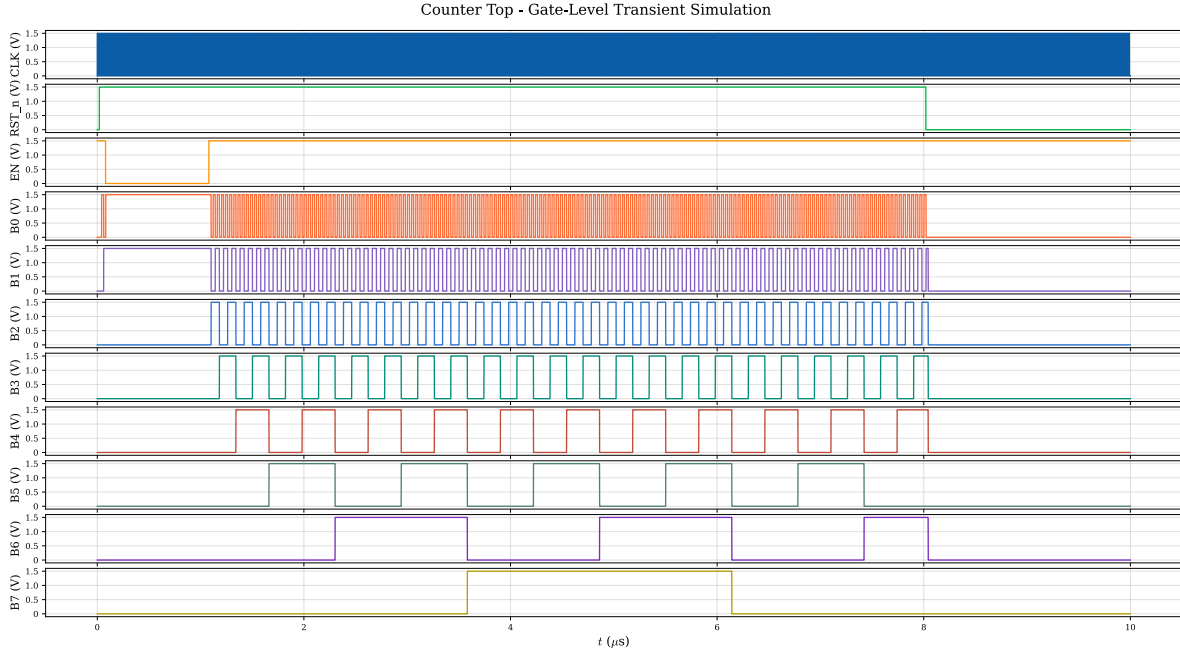


Figure 7: Plotted Xschem simulation results of `counter_top_tb_tran` from `testbenches/xschem/plot_simulations/plot_counter_top.py`.

## 4.9 Parasitic Extraction (PEX)

The XSPICE model above is a *gate-level* view of the macro: every standard cell becomes an XSPICE primitive driven by its Liberty delay, and the wiring is ideal. Parasitic extraction gives the *transistor-level* view instead. It reads the hardened layout `final/gds/counter_top.gds` and writes a SPICE netlist into `netlist/pex/` in which every standard cell is flattened to transistors and the wiring carries the resistances and capacitances measured from the layout.

```
make magic-pex                                # full-RC, the default EXT_MODE=3
make magic-pex EXT_MODE=1                      # C-decoupled, fastest
make klayout-pex                              # the same with kplex, which runs Magic internally
make magic-pex EXT_MODE=3 THRESHOLD=5000 MINRES=500 MINDELAY=2 # full-RC with custom extremes
```

Both targets read `final/gds/<CELL>.gds`, so `make build-top` (or at least `make copy-final`) has to have run first, and both abort with a clear message if it is missing. They start by running `make symbol-pex`, which derives `schematic/xschem/counter_top_pex.sym` from `counter_top.sym` so the PEX symbol can never go stale, and they finish by reordering the extracted `.subckt` ports to that symbol's pin order with `sak-pin-reorder.py` and verifying them with `scripts/check_pex_ports.py`. For `counter_top`, the full-RC default extracts 4401 transistors and roughly 2100 capacitors and 4600 resistors. Magic's `extresist` step is not

deterministic, so the exact R and C counts move slightly from run to run while the transistor count stays the same.

The transient testbench already `.includes` the extracted netlist next to the XSPICE model, and it carries two spare DUT instances above the schematic, `x2` of `counter_top.sym` and `x3` of `counter_top_pex.sym`, both parked with `spice_ignore=true`. Swapping the wired-up `x1` for `x3` turns the run into a post-layout simulation.

#### Warning

A post-layout run simulates every transistor of the macro. In the IIC-OSIC-TOOLS container, 60 ns of transient take about 80 s, while the full 10  $\mu$ s gate-level XSPICE run of the same testbench finishes in under 2 s. The XSPICE model stays the right tool for functional runs. Use the extracted netlist on short, targeted stimuli to check timing and signal integrity.

#### Note

The testbench uses looser Ngspice tolerances than the analog macro (`reltol=1e-3` `abstol=1e-12` `gmin=1e-12` `rshunt=1e14`). Full-RC extraction splits the nets into fragments, and some of the resulting nodes have no DC path to ground, so with the analog macro's settings the post-layout run aborts at the initial timepoint with `Timestep too small`. The gate-level XSPICE run gives the same result either way.

## 4.10 Lint, Build, Verify and Simulate Everything

The full lint, build, verification, and simulation chain is run with the following command.

```
make all
```

This calls `lint-verilog-all`, `build-fpga`, `build-top`, `magic-pex`, and `sim-all` in sequence. The simulations run after the build, so the gate-level simulations run on the netlists, the XSPICE model and the extracted netlist produced by this build. `klayout-pex` is commented out in the recipe, since it is a second full extraction that nothing in the flow consumes. DRC and LVS are already covered by the LibreLane verification steps, so no extra verification target is needed for the digital macro.

## 5 Analog Macro Implementation

The `inverter` macro is the analog reference macro of the template. It is a 4-channel CMOS inverter, drawn at transistor level, and verified with a comprehensive verification chain:

- Layout vs. Schematic ([LVS](#)) with KLayout or Magic + Netgen,
- Design Rule Checking ([DRC](#)) with KLayout or Magic,
- Parasitic Extraction ([PEX](#)) with KLayout or Magic,
- [Monte Carlo](#) simulation of process and mismatch variations with CACE.

Two instances are used in the chip core. One is used as a 4-channel digital inverter (`inverter1`), and the other as a 2-channel analog inverter (`inverter2`). See the [chip specifications](#) for the full picture.

#### Note

An inverter can act as a digital circuit (when the input is driven by a digital signal and the output is switching between the supply rails). Here, we characterise the inverter as an analog circuit, meaning that we are interested in its continuous voltage transfer characteristic and its small-signal voltage gain.

## 5.1 Directory Structure

The inverter design files are located in [macros/inverter/](#), and the relevant folders are:

| Folder                              | Content                                                                                                                                                                                                        |
|-------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <a href="#">schematic/xschem/</a>   | Xschem schematics and symbols ( <code>inverter.sch</code> , <code>inverter.sym</code> , <code>inverter_top.sch</code> , <code>inverter_top.sym</code> , and the generated <code>inverter_top_pex.sym</code> ). |
| <a href="#">layout/</a>             | KLayout GDS files ( <code>*.klay.gds</code> ) and the exported <code>inverter_top.gds</code> .                                                                                                                 |
| <a href="#">testbenches/xschem/</a> | Transient, ac open-loop, dc input sweep, and top-level transient testbenches.                                                                                                                                  |
| <a href="#">netlist/</a>            | <code>schematic/</code> (CDL for KLayout LVS / SPICE for Magic + Netgen LVS), <code>layout/</code> (extracted for LVS), <code>pex/</code> (SPICE).                                                             |
| <a href="#">scripts/</a>            | Sizing notebooks, layout rendering.                                                                                                                                                                            |
| <a href="#">verification/</a>       | <code>drc/</code> , <code>lvs/</code> , and <code>cace/</code> (process, mismatch, supply sweeps).                                                                                                             |
| <a href="#">final/</a>              | Build deliverables for the top-level integration (LEF, LIB, Verilog stub, GDS).                                                                                                                                |
| <a href="#">render/</a>             | Macro renders for documentation.                                                                                                                                                                               |

#### Tip

Read the [inverter README](#) first to get familiar with the folder layout and the available Makefile targets. Most of the commands below are documented there as well.

## 5.2 Schematic-Level Simulation

Go to the Xschem testbench folder and start Xschem.

```
cd macros/inverter/testbenches/xschem
xschem
xschem <testbench_name>.sch # load a testbench schematic directly at startup
```

Four testbenches are available in [testbenches/xschem/](#).

- [inverter\\_tb\\_tran.sch](#), single-channel transient response (see Figure 8),
- [inverter\\_tb\\_ac\\_ol.sch](#), open-loop ac characterisation (see Figure 9),
- [inverter\\_tb\\_dc\\_vout.sch](#), dc input sweep (see Figure 10),
- [inverter\\_top\\_tb\\_tran.sch](#), top-level transient on `inverter_top` (see Figure 11).

Load a testbench in Xschem by pressing **Ctrl + o**, then hold **Ctrl** and click the **Simulate** button in the schematic to launch the simulator. Once the simulation has finished, hold **Ctrl** and click the **Load waves** arrow to load the results into the integrated waveform viewer. The following figures show the testbench schematics and selected plotted results.

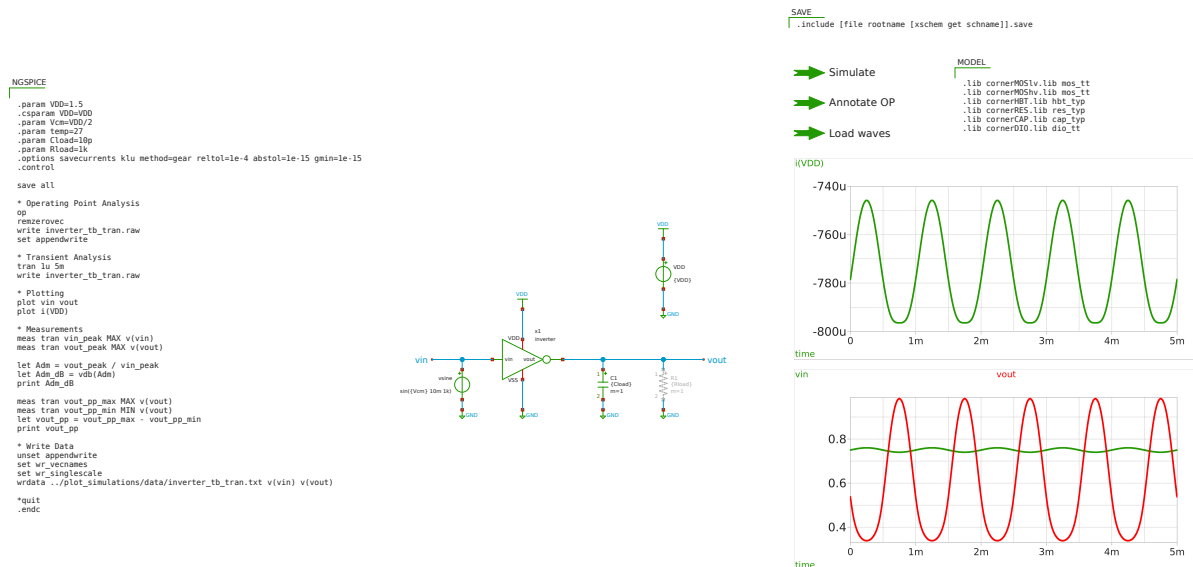


Figure 8: Xschem testbench `inverter_tb_tran` (single-channel transient response on `inverter`).

Non-interactive runs and post-processing are also available through the Makefile.

```

NGSPICE
.param VDD=1.5
.param Vcm=VDD/2
.param temp=27
.param Cload=10p
.param Rload=1k
.options savecurrents klu method=gear reitol=1e-4 abstol=1e-15 gmin=1e-15
.control

save all

set wr_vecnames
set wr_singlescale

* User Constants
let f_min = 0.1
let f_max = 100
let fdc = 1

* Operating Point Analysis
op
remzerovec
write inverter_tb_ac_ol.raw
set appendwrite

* AC Analysis
ac dec 101 50const.f_min 50const.f_max
remzerovec
write inverter_tb_ac_ol.raw

* Plotting
let AoI = v(vout)/v(vin)
let AoI_db = vdb(AoI)
let AoI_arg = 180/Pi*cphase(AoI)
plot AoI_db ylabel 'Magnitude'
plot AoI_arg ylabel 'Phase'
plot AoI_db AoI_arg ylabel 'Magnitude, Phase'

* Measurements
* DC open-loop gain
meas ac Aol_db find AoI_db when frequency = fdc
print Aol_db

* 3dB cut-off frequency
let AoI_fc = Aol_db - 3
meas ac fc find frequency when AoI_db = AoI_fc
print fc

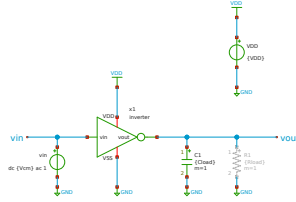
* Unity Gain Bandwidth (Aol=1 or AoI_db = 0dB)
meas ac UGB when AoI_db=0 fat=1
print UGB

* Phase Margin at Aol=1 or AoI_db = 0dB
meas ac arg_0dB find AoI_arg when AoI_db=0
let PM = 180-abs(arg_0dB)
print PM

* Write Data
unset appendwrite
set wr_vecnames
set wr_singlescale
wdata ../plot_simulations/data/inverter_tb_ac_ol.txt v(AoI_db) v(AoI_arg)

*quit
.endc

```



```

SAVE
[.include [file rootname [xschem get schname]].save

```

- ➡ Simulate
- ➡ Annotate OP
- ➡ Load waves

```

MODEL
.lib cornerMOSlv.lib mos_tt
.lib cornerMOSlv.lib mos_tt
.lib cornerMOSlv.lib mos_tt
.lib cornerRES.lib res_typ
.lib cornerCAP.lib cap_typ
.lib cornerDIO.lib dio_tt

```

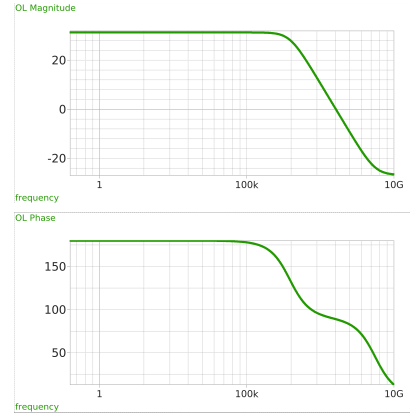


Figure 9: Xschem testbench inverter\_tb\_ac\_ol (open-loop ac characterisation on inverter).

```

NGSPICE
.param VDD=1.5
.cparam VDD=VDD
.param Vcm=VDD/2
.cparam Vcm=Vcm
.param temp=27
.param Cload=10p
.param Rload=1k
.options savecurrents klu method=gear reitol=1e-3 abstol=1e-15 gmin=1e-15
.control

save all

* Operating Point Analysis
op
remzerovec
write inverter_tb_dc_vout.raw
set appendwrite

* DC Sweep
dc Vgsp 0 50VDD 1m
remzerovec
write inverter_tb_dc_vout.raw
set appendwrite

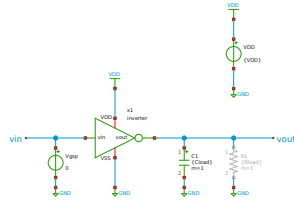
* Plotting
plot vin vout

* Measurement
meas dc Vgsp_at_Vcm when vout=Vcm
print Vgsp_at_Vcm

* Write Data
unset appendwrite
set wr_vecnames
set wr_singlescale
wdata ../plot_simulations/data/inverter_tb_dc_vout.txt v(vin) v(vout)

*quit
.endc

```



```

SAVE
[.include [file rootname [xschem get schname]].save

```

- ➡ Simulate
- ➡ Annotate OP
- ➡ Load waves

```

MODEL
.lib cornerMOSlv.lib mos_tt
.lib cornerMOSlv.lib mos_tt
.lib cornerMOSlv.lib mos_tt
.lib cornerRES.lib res_typ
.lib cornerCAP.lib cap_typ
.lib cornerDIO.lib dio_tt

```

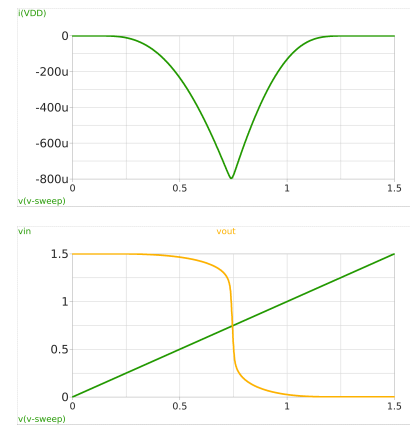


Figure 10: Xschem testbench inverter\_tb\_dc\_vout (dc input sweep on inverter).

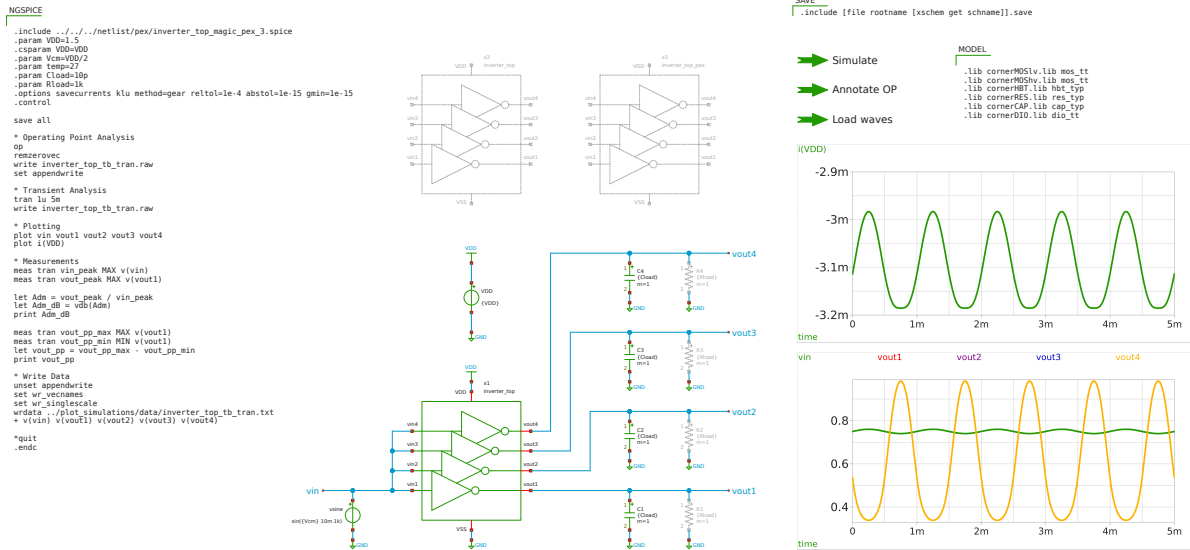


Figure 11: Xschem top-level testbench `inverter_top_tb_tran` (transient on `inverter_top`).

```
cd macros/inverter
make sim-xschem # run the default testbench (inverter_top_tb_tran)
make sim-xschem TB=<testbench_name> # run another testbench non-interactively
make sim-view-xschem SCRIPT=plot_inverter # plot results of inverter with Python
make sim-view-xschem SCRIPT=plot_inverter_top # plot results of inverter_top with Python
```

`sim-xschem` runs in batch mode: it netlists the testbench with `xschem netlist` and then calls `ngspice -b` directly, so `make` blocks until the run finishes and sees its exit status. The testbench is selected with `TB=<testbench_name>` and defaults to `<CELL>_tb_tran`. Since the run is headless, the `plot` commands in the testbench's `.control` block are a no-op. Every testbench exports its results with `wrdata` to `testbenches/xschem/plot_simulations/data/` instead.

The Makefile target `sim-view-xschem` calls the Python plotters in `testbenches/xschem/plot_simulations/`, selected with the `SCRIPT` variable (default: `plot_<CELL>`), to parse the raw simulation output and plot the results with Matplotlib. The following figures show some of these plotted results.

#### 💡 Tip

The full simulation suite, including all four testbenches plus the CACE flow (see Section 5.4), is triggered with `make sim-all`.

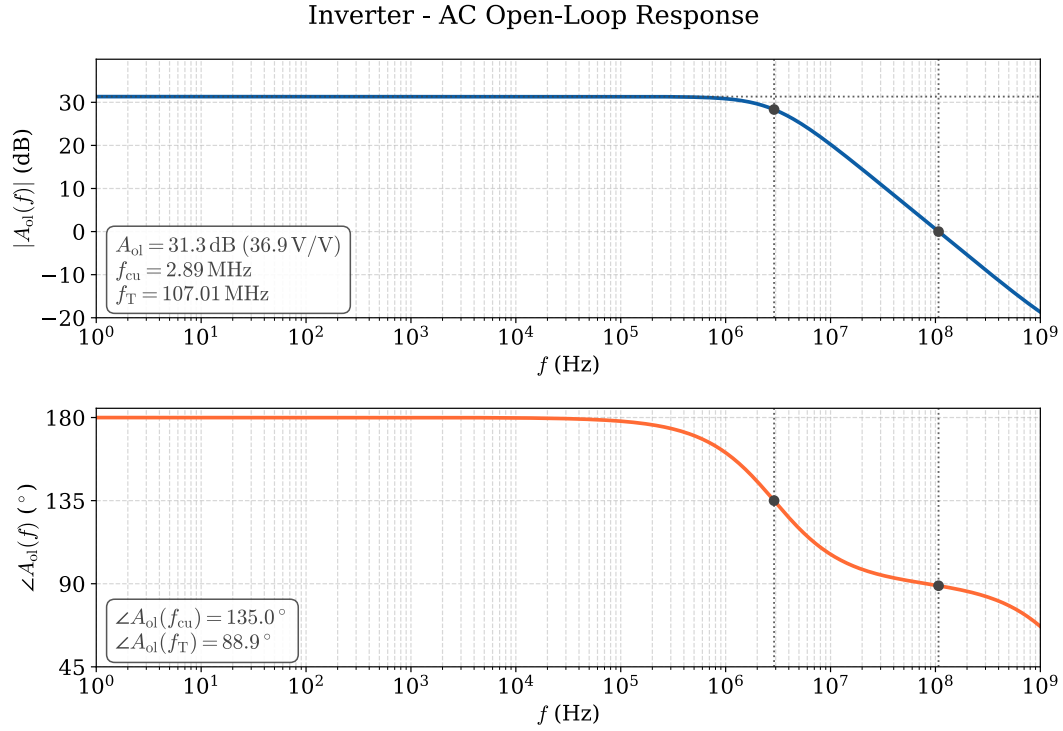


Figure 12: Plotted simulation result of `inverter_tb_ac_ol`.

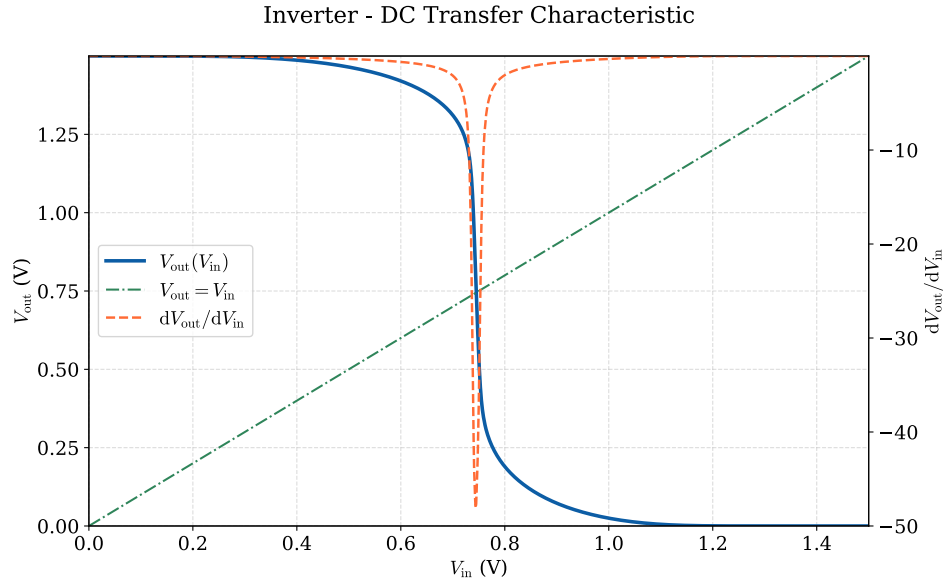


Figure 13: Plotted simulation result of `inverter_tb_dc_vout`.

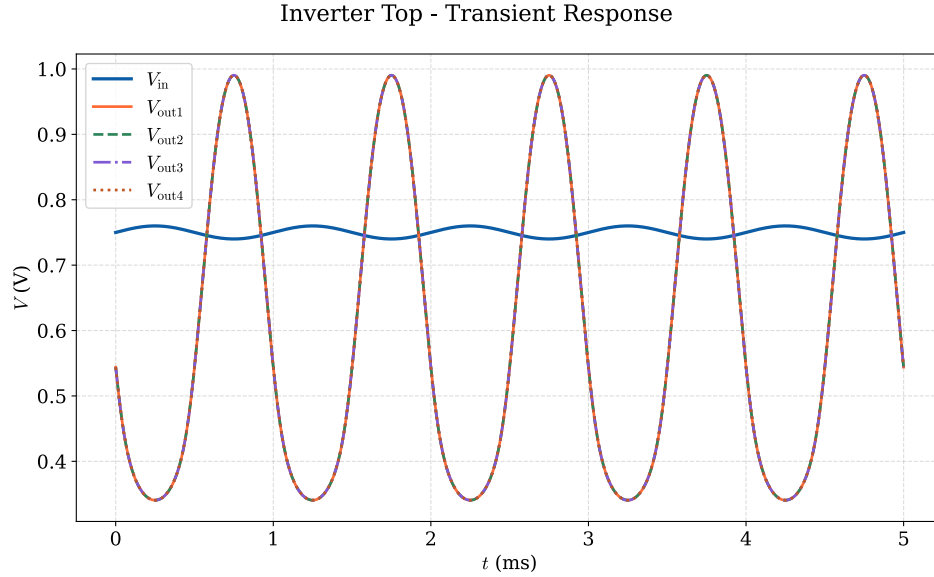


Figure 14: Plotted simulation result of `inverter_top_tb_tran`.

### 5.3 Schematic-Level Drawing

To inspect or modify the underlying circuit of a symbol, mark the symbol and press **e** to descend into the underlying circuit, then **Ctrl + e** to go back up. To inspect or modify the graphical symbol of a device, mark the symbol and press **i** to descend into the symbol drawing, then **Ctrl + e** to go back up again.

#### Tip

Since Xschem is a text-based schematic editor, you can also open `*.sch` and `*.sym` files in a text editor and modify them there. This is especially useful for search-and-replace operations.

Figure 15 shows the symbol of the `inverter` cell.

Figure 16 displays the schematic of the `inverter` cell with additional small-signal parameters for the NMOS and PMOS transistors at the chosen operating point. To view these small-signal parameters, hold **Ctrl** and click **Annotate OP** after the simulation has finished. Afterwards, descend into the inverter schematic.

### 5.4 Process Variation and Mismatch (CACE)

**CACE** drives Ngspice via a characterisation YAML file ([verification/cace/inverter.yaml](#))



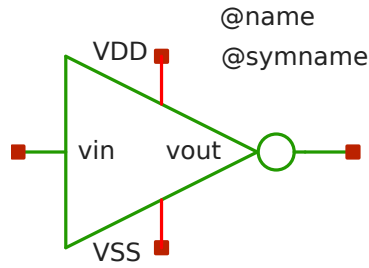


Figure 15: Xschem inverter symbol.

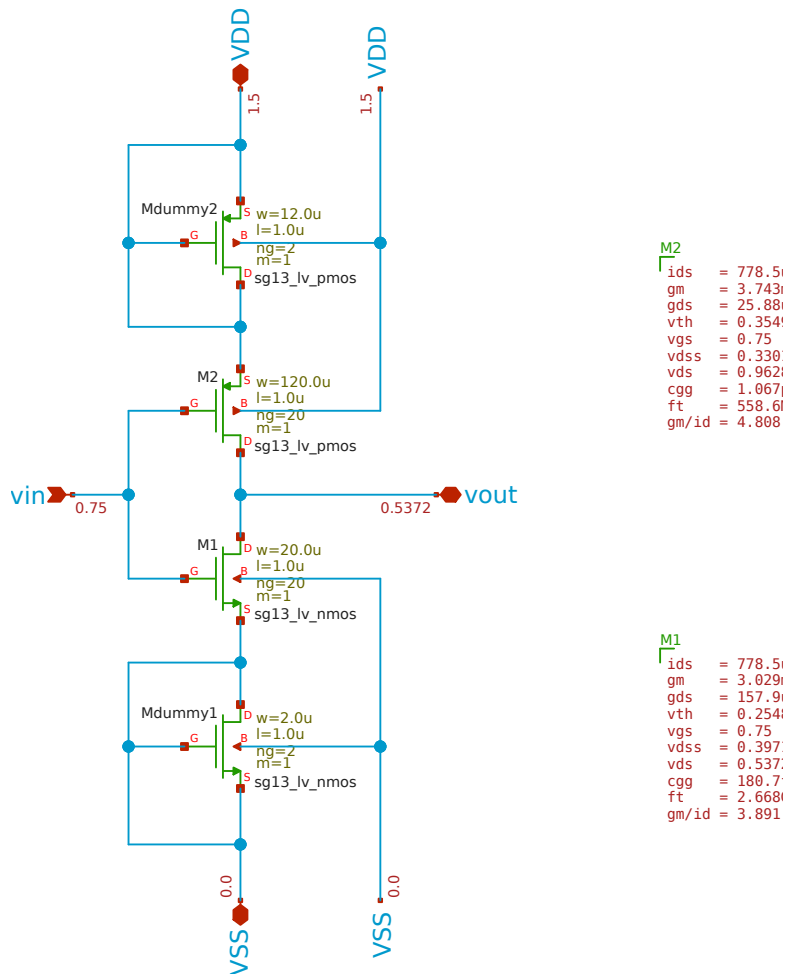


Figure 16: Xschem inverter schematic with dummy transistors and OP annotations.

to run Monte Carlo simulations and produce process and mismatch variation plots.

```
make sim-cace
```

Plots are written to `verification/cace/results/inverter/`. This step is mainly relevant for advanced designers who want to characterise the analog macro across PVT corners. The following figures show the results of the CACE simulations for 200 iterations.

Figure 17 shows the process (left) and mismatch (right) variations of the dc open-loop gain  $A_{ol}$ , while Figure 18 plots its variation across several process corners at one fixed supply voltage.

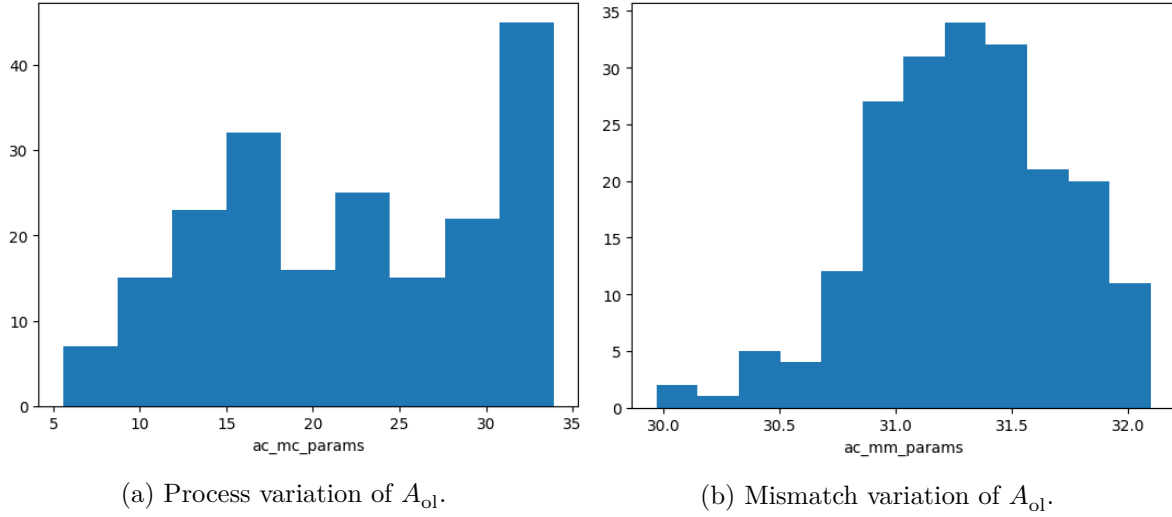


Figure 17: CACE process (left) and mismatch (right) variations of  $A_{ol}$ .

Figure 19 shows the process (left) and mismatch (right) variations of the open-loop lower cut-off frequency  $f_{cu}$ , while Figure 20 plots its variation across several process corners at one fixed supply voltage.

#### **i** Note

Note that the above process and mismatch variation results are poor for an analog circuit. Also, the open-loop gain varies significantly across corners. This is because the example is only a simple inverter without a compensation technique. For a practical single-ended inverter amplifier, one would at least add a feedback resistor from output to input to stabilise the operating point and therefore the gain across corners. If you want to see improved variation results, you can try adding a feedback resistor in the `inverter.sch` schematic and running the CACE flow again.

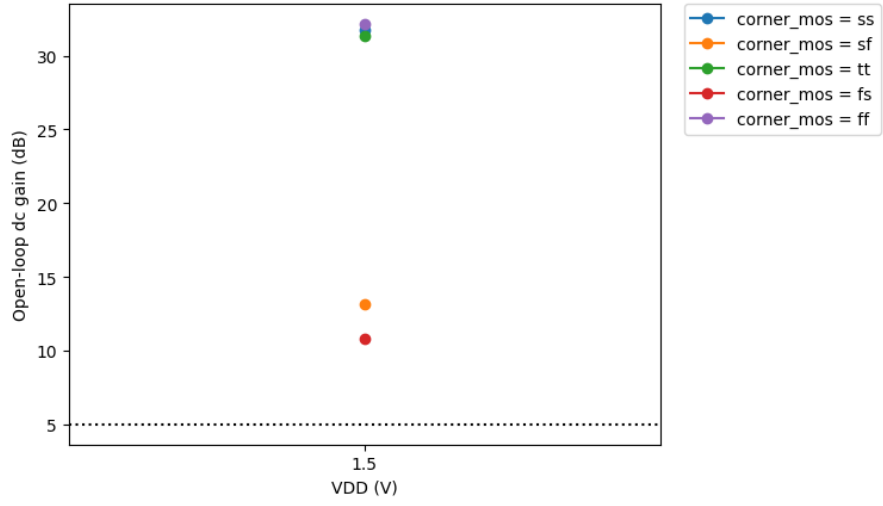


Figure 18: CACE variation of  $A_{ol}$  across several process corners.

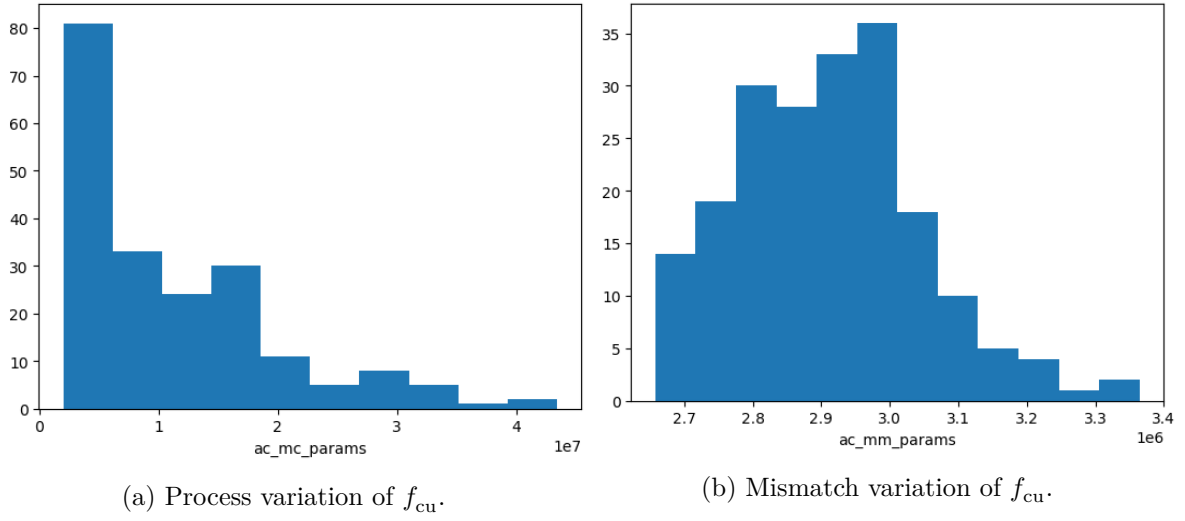


Figure 19: CACE process (left) and mismatch (right) variations of  $f_{cu}$ .

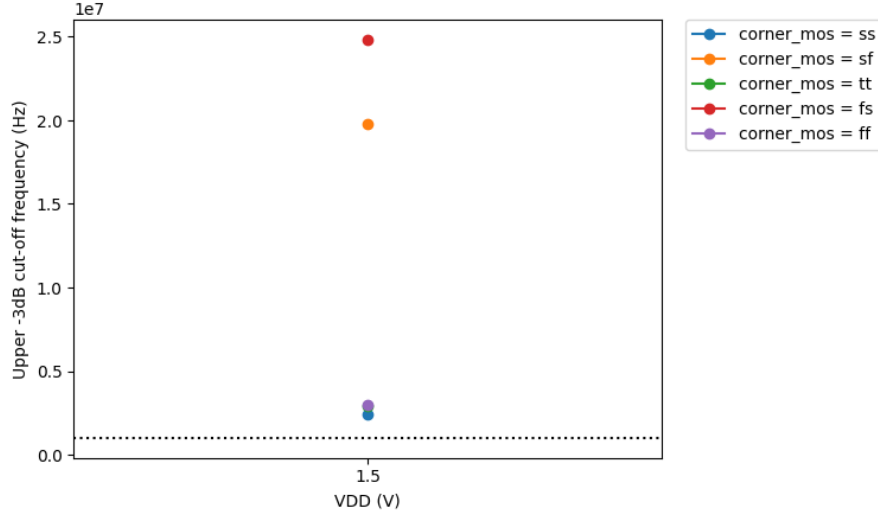


Figure 20: CACE variation of  $f_{cu}$  across several process corners.



Tip

In the `inverter.yaml` characterisation file, more parameter sweeps can be added, for example, across temperature, supply voltages or load capacitance.

## 5.5 Transistor Sizing

During simulation, it could happen that some specifications are not fulfilled and that the inverter must be resized. In this tutorial, the inverter is sized with the  $g_m/I_D$  method (see [Transistor Sizing Using  \$g\_m/I\_D\$  Methodology](#) for theoretical background). Instead of relying on textbook square-law equations, the method looks up the actual device behaviour from pre-characterised SPICE simulations, parameterised by  $g_m/I_D$ . An overview of these techsweeps can be found in the folder [doc/sizing/](#). The chosen  $g_m/I_D$  point sets the inversion level of the transistor and therefore the trade-off between intrinsic gain (high  $g_m/I_D$ , weak inversion) and bandwidth (low  $g_m/I_D$ , strong inversion). An overview of the  $g_m/I_D$  trade-offs can be found [here](#). For each chosen operating point the lookup tables return all relevant figures of merit ( $I_D/W$ ,  $g_m/g_{ds}$ ,  $g_m/C_{gg}$ ,  $S_{TH}$ , etc.) at the actual bias condition, which is then used to compute  $W$  for a given drain current.

The full sizing flow for the inverter is implemented in the Jupyter notebook [scripts/sizing/sizing\\_inverter.ipynb](#). The notebook uses `pygmid` to query the pre-characterised ihp-sg13g2 NMOS/PMOS lookup tables ([sg13g2\\_lv\\_nmos.mat](#) and [sg13g2\\_lv\\_pmos.mat](#)). An overview of the  $g_m/I_D$  MOSFET characterisation procedure can be found [here](#).

The Jupyter notebook `sizing_inverter.ipynb` runs through the following steps:

1. Define the design targets:  $V_{DD} = 1.5 \text{ V}$ ,  $V_{cm,in} = V_{cm,out} = V_{DD}/2 = 0.75 \text{ V}$ ,  $I_{out} = 0.75 \text{ mA}$ ,  $C_{load} = 10 \text{ pF}$  and  $L = 1 \text{ m}$ .  $L$  is chosen longer than the minimum to increase intrinsic gain and reduce mismatch, but at the cost of increased area and reduced bandwidth.
2. Look up  $g_m/I_D$ ,  $g_m/g_{ds}$ ,  $g_m/C_{gg}$  and  $I_D/W$  based on  $L$  and the operating-point voltages.
3. Compute the required widths  $W = I_D/(I_D/W)$  for NMOS and PMOS, and derive the resulting  $W_{PMOS}/W_{NMOS}$  ratio.
4. Round the continuous widths to a practical finger configuration with equal finger counts for NMOS and PMOS (for better layout matching) and recompute the small-signal parameters with the final widths.
5. Estimate the open-loop gain  $A_{ol} = -g_{m,tot}/g_{ds,tot}$  and the output resistance  $R_{out} = 1/g_{ds,tot}$  of the complementary inverter stage.

The resulting device geometry is summarised in Table 1.

Table 1: Sized device geometry of the complementary inverter.

| De-<br>vice   | $L$ (m) | Fingers (NF) | $W$ / Finger (m) | Total $W$ (m) |
|---------------|---------|--------------|------------------|---------------|
| NMOS<br>$M_1$ | 1.0     | 20           | 1.0              | 20.0          |
| PMOS<br>$M_2$ | 1.0     | 20           | 6.0              | 120.0         |

With these final widths the notebook computes the small-signal parameters listed in Table 2. As a reference, compare these values with the OP annotations in Figure 16.

Table 2: Calculated small-signal parameters with final widths.

| Parameter                                             | NMOS<br>$M_1$ | PMOS<br>$M_2$ | Inverter (NMOS    PMOS) |
|-------------------------------------------------------|---------------|---------------|-------------------------|
| Drain current $I_D$ (mA)                              | 0.89          | 0.78          | —                       |
| Transconductance efficiency $g_m/I_D$<br>( $V^{-1}$ ) | 3.76          | 3.76          | —                       |
| Inversion level                                       | strong        | strong        | —                       |
| Transconductance $g_m$ (mS)                           | 3.35          | 2.92          | —                       |
| Output conductance $g_{ds}$ (S)                       | 116           | 29            | —                       |
| Gate capacitance $C_{gg}$ (fF)                        | 154.32        | 824.86        | —                       |
| Transit frequency $f_T = g_m/(2\pi C_{gg})$<br>(GHz)  | 2.91          | 0.54          | —                       |

| Parameter                                                          | NMOS<br>$M_1$ | PMOS<br>$M_2$ | Inverter (NMOS    PMOS) |
|--------------------------------------------------------------------|---------------|---------------|-------------------------|
| Thermal noise PSD $S_{\text{TH}}$ at 1 Hz<br>(pV <sup>2</sup> /Hz) | 38.95         | 73.98         | –                       |
| Flicker corner frequency $f_{\text{co}}$ (MHz)                     | 3.87          | 2.28          | –                       |
| $g_{\text{m,tot}} = g_{\text{m,NMOS}} + g_{\text{m,PMOS}}$ (mS)    | –             | –             | 6.28                    |
| $g_{\text{ds,tot}} = g_{\text{ds,NMOS}} + g_{\text{ds,PMOS}}$ (S)  | –             | –             | 145                     |

From these values, the open-loop dc gain  $A_{\text{ol}}$ , the output resistance  $R_{\text{out}}$ , the open-loop cut-off frequency  $f_{\text{cu}}$ , and the unity-gain frequency  $f_{\text{T}}$  with a dominant capacitive load  $C_{\text{load}} = 10$  pF can be estimated as

$$|A_{\text{ol}}| = \frac{g_{\text{m,tot}}}{g_{\text{ds,tot}}} = \frac{6.28 \text{ mS}}{145 \text{ S}} = 43.2 = 32.72 \text{ dB},$$

$$R_{\text{out}} = \frac{1}{g_{\text{ds,tot}}} = \frac{1}{145 \text{ S}} = 6.89 \text{ k},$$

$$f_{\text{cu}} = \frac{1}{2\pi R_{\text{out}} C_{\text{load}}} = \frac{1}{2\pi \cdot 6.89 \text{ k} \cdot 10 \text{ pF}} = 2.31 \text{ MHz},$$

$$f_{\text{T}} = \frac{g_{\text{m,tot}}}{2\pi C_{\text{load}}} = \frac{6.28 \text{ mS}}{2\pi \cdot 10 \text{ pF}} = 100 \text{ MHz}.$$

Table 3 compares these hand-calculated values against the open-loop ac simulation result from [inverter\\_tb\\_ac\\_ol](#) (see Figure 12).

Table 3: Calculated vs. simulated open-loop characteristics of the inverter.

| Quantity                                    | $g_{\text{m}}/I_{\text{D}}$ calculation | ac open-loop simulation |
|---------------------------------------------|-----------------------------------------|-------------------------|
| Open-loop dc gain $A_{\text{ol}}$           | 32.72 dB                                | 31.33 dB                |
| Open-loop cut-off frequency $f_{\text{cu}}$ | 2.31 MHz                                | 2.89 MHz                |
| Unity-gain frequency $f_{\text{T}}$         | 100 MHz                                 | 107 MHz                 |

#### Tip

The notebook is fully parametric. Changing  $L$ , the target  $g_{\text{m}}/I_{\text{D}}$  or  $I_{\text{out}}$  and re-running the cells immediately produces a new sizing point, so it can be used as a starting template for sizing other simple analog blocks with the ihp-sg13g2 lookup tables. Try it out yourself

by decreasing the transistor length to  $L = 0.5\text{ m}$  and see how the small-signal parameters and the resulting open-loop gain and cut-off frequency change. What would you expect based on the trade-offs above?

## 5.6 Inspect / Edit the Layout

After the schematic is finalised and the design targets are met, the layout can be created. The layout of the `inverter` cell is hierarchical, and the top-level cell `inverter_top` contains four instances of the `inverter` cell. Around the 4-channel inverter layout, a power ring is drawn on `Metal5` and `TopMetal1` layers, which is connected to the `VDD` and `VSS` pins of the four inverters. This power ring is required for top-level integration, connecting the macro power to the chip's PDN.

The layout is created and edited in KLayout. The container provides the `ke` alias for `klayout -e` (edit mode). With the following command, the GDS of the 4-channel inverter is opened in KLayout as shown in Figure 21. With the keys 0, 1, 2, ..., 9 you can view different layers as explained in the [KLayout cheatsheet](#) or in the [klayout-layer-shortcuts](#) plugin.

```
ke layout/inverter_top.klay.gds
```

After modifying the layout, export a clean `.gds` file (for example `inverter_top.gds`) by pressing `File > Export Layout For Tapeout`. All verification flows (DRC, LVS, and PEX with both KLayout and Magic) accept either `.gds` or `.klay.gds`, while the build targets (LEF export, GDS copy, and rendering) use the exported `.gds`.

KLayout also supports a 2.5D viewer, which can be used to inspect the layout. To open the 2.5D viewer, press `SG13G2 PDK > BEOL 2.5D Viewer`. Figure 22 shows the `inverter_top` layout in the 2.5D viewer.

## 5.7 DRC, LVS, and PEX

The inverter macro can be verified with two verification flows, either with KLayout or with Magic + Netgen. Each flow covers DRC, LVS, and PEX. Magic + Netgen provides only non-interactive command-line flows, while KLayout also has interactive GUI-based DRC and LVS.

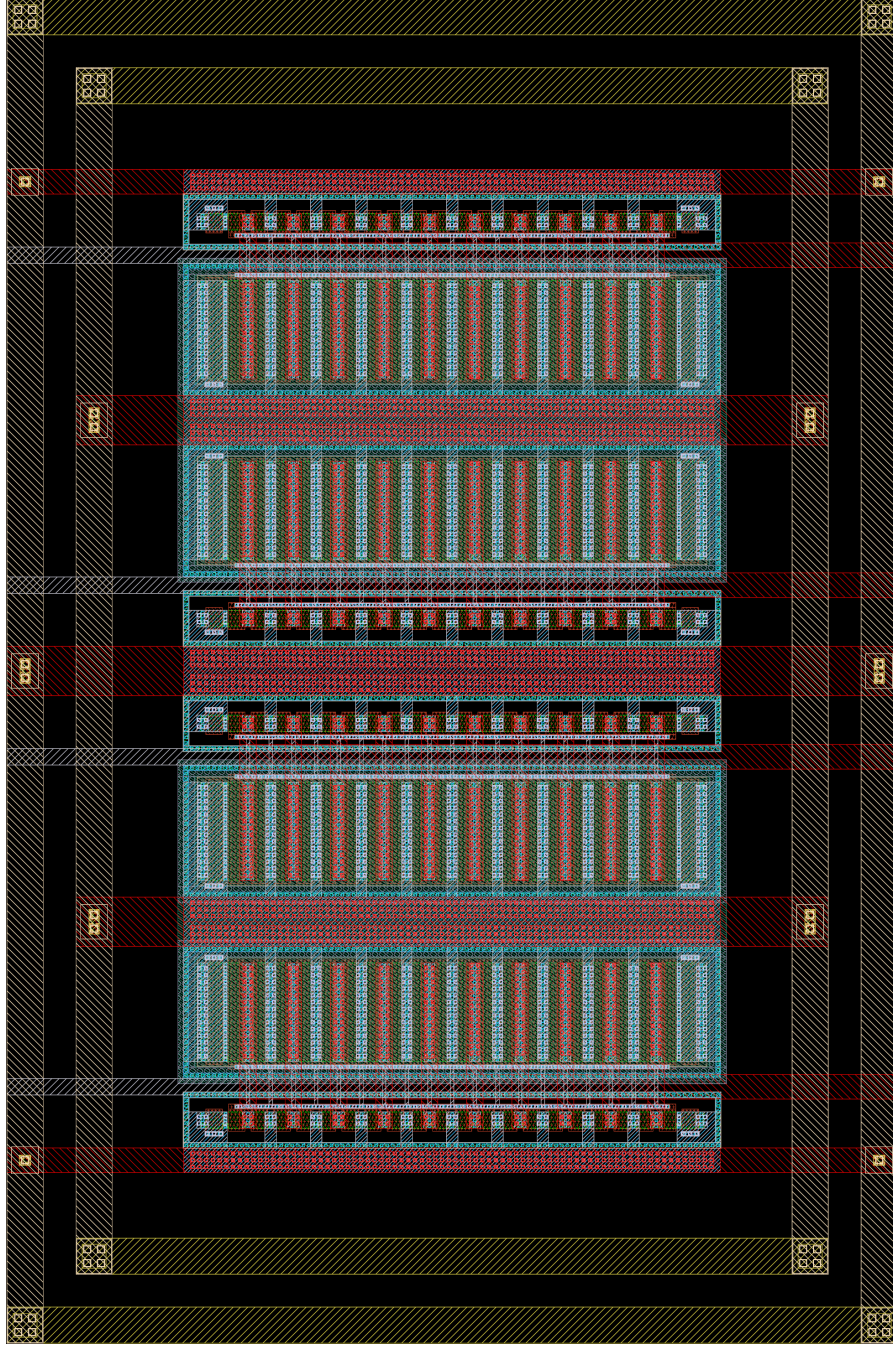


Figure 21: inverter\_top layout in KLayout.



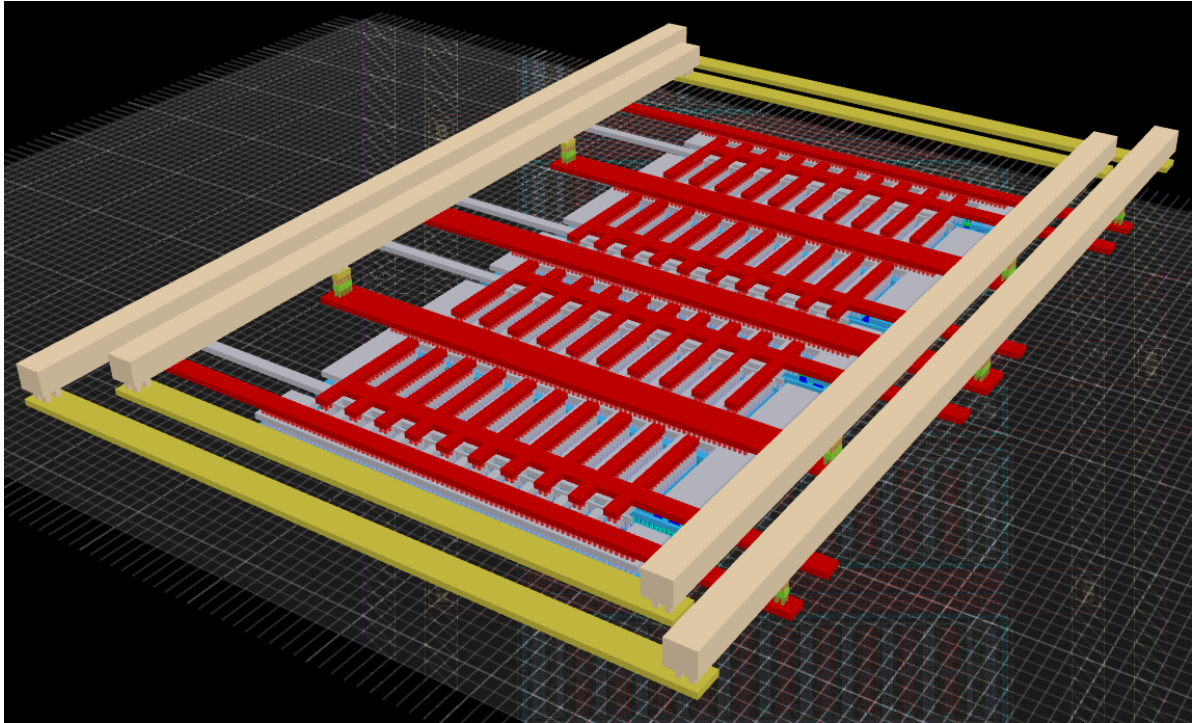


Figure 22: inverter\_top layout in KLayout 2.5D viewer.

### 5.7.1 Design Rule Check

The non-interactive DRC can be run with the following Makefile targets (both use `sak-drc.sh`). The `CELL` variable can be set to run DRC on a specific cell, or it defaults to `inverter_top` if not set. The `DRC_LEVEL` variable selects the KLayout DRC level: `precheck`, `macro` (default), or `regular`. It is ignored by `magic-drc`, since Magic has no selectable rule decks and always runs the full rule set compiled into the PDK's Magic tech file.

| Check                     | precheck | macro ( <i>default</i> ) | regular |
|---------------------------|----------|--------------------------|---------|
| FEOL + BEOL core rules    |          |                          |         |
| Off-grid / angle          | —        |                          |         |
| Zero-area / geometry      | —        |                          |         |
| Pin / label               | —        |                          |         |
| Recommended / extra rules | —        | —                        |         |
| Density (full-chip fill)  | —        | —                        |         |
| Antenna                   | —        | —                        |         |

```
make klayout-drc [CELL=<cellname>] [DRC_LEVEL=<precheck|macro|regular>] # default CELL=inverter_top
make magic-drc [CELL=<cellname>] # default CELL=inverter_top
```

Reports are written into per-cell run folders (`<CELL>.magic.drc/` for Magic, `<CELL>.klayout.drc/` for KLayout) inside [verification/drc/](#). The run folders are wiped at the start of each run, so they always reflect the latest run only.

#### Tip

In KLayout, a non-interactive DRC run can be launched from the GUI by pressing **F9**. The DRC options can be modified by pressing **F10**.

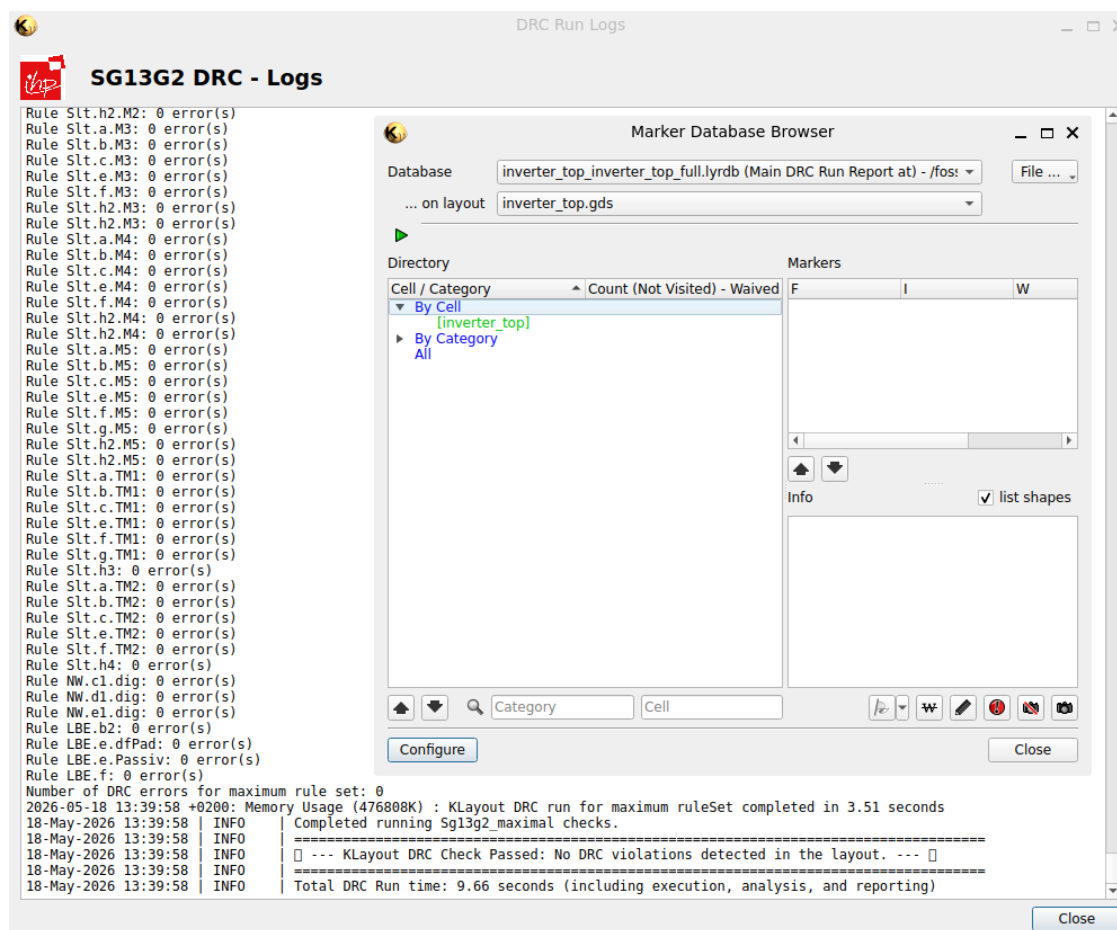


Figure 23: KLayout DRC output windows.

## 5.7.2 Layout-Versus-Schematic

The non-interactive LVS can be run with the following Makefile targets. The `CELL` variable can be set to run LVS on a specific cell, or it defaults to `inverter_top` if not set.

```
make klayout-lvs [CELL=<cellname>] # default CELL=inverter_top, uses CDL netlist exported from klayout
make magic-lvs [CELL=<cellname>]   # default CELL=inverter_top, uses SPICE netlist exported from magic
```

The schematic netlist is exported automatically by the matching `*-lvs-netlist` sub-target and moved to `netlist/schematic/`. The extracted layout netlist is moved to `netlist/layout/` and the reports are written into per-cell run folders (`<CELL>.magic.lvs/` for Magic + Netgen,

<CELL>.klayout.lvs/ for KLayout) inside [verification/lvs/](#). The run folders are wiped at the start of each run, so they always reflect the latest run only.

💡

**Tip**

In KLayout, a non-interactive LVS run can be launched from the GUI by pressing **F11**. The LVS options can be modified by pressing **F12**, where the schematic netlist `inverter_top_klayout.cdl` has to be selected manually. Set `deep` as the run mode.

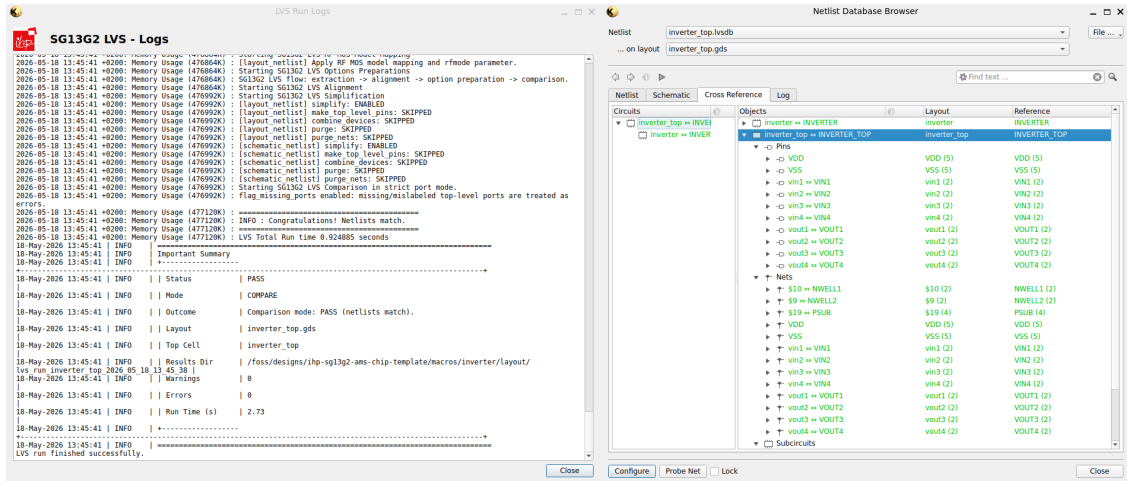


Figure 24: KLayout LVS output windows.

### 5.7.3 Parasitic Extraction

The non-interactive PEX can be run with the following Makefile targets. The `CELL` variable can be set to run PEX on a specific cell, or it defaults to `inverter_top` if not set. The `EXT_MODE` variable can be set to choose between different extraction modes,

- `EXT_MODE=1` for C-decoupled mode,
- `EXT_MODE=2` for C-coupled mode,
- `EXT_MODE=3` for full-RC extraction.

If not set, it defaults to `EXT_MODE=3`.

For full-RC extraction (`EXT_MODE=3`), `magic-pex` also exposes the `sak-pex.sh` extresist tuning parameters (ignored in `EXT_MODE=1/2`):

- `THRESHOLD` - extresist threshold in mOhm (`-t`, default 10000 = 10 Ohm),
- `MINRES` - extresist minimum resistance in mOhm (`-r`, default 1000 = 1 Ohm),
- `MINDELAY` - extresist minimum delay in ps (`-y`, default 1; 0 = gate by resistance).

```
make klayout-pex [CELL=<cellname>] [EXT_MODE=<1|2|3>] # default CELL=inverter_top, default
make magic-pex [CELL=<cellname>] [EXT_MODE=<1|2|3>] [THRESHOLD=<m0hm>] [MINRES=<m0hm>] [MINDI
```

PEX writes a SPICE netlist into `netlist/pex/`. The subcircuit is named `<CELL>_pex`, and the pin order is rearranged to match the Xschem PEX symbol (`inverter_top_pex.sym`).

That PEX symbol is generated, not maintained by hand. Both PEX targets start by running `make symbol-pex`, which copies `schematic/xschem/<CELL>.sym` to `schematic/xschem/<CELL>_pex.sym` and changes `type=subcircuit` into `type=primitive`. Everything else is inherited, above all the pin order that `sak-pin-reorder.py` sorts the extracted netlist to. The `primitive` type keeps Xschem from descending into a schematic of the same name, so the instance is netlisted as it stands and its subcircuit comes from the included PEX netlist instead. Because the symbol is rebuilt before every extraction, it cannot go stale when a pin is added, removed or renamed, and it can be built on its own with `make symbol-pex [CELL=<cellname>]`. Edit `<CELL>.sym`, never the generated `<CELL>_pex.sym`.

#### Note

Every symbol in this template carries `spectre_format="@name ( @pinlist ) @symname"` next to its `format` line. Xschem writes that line itself whenever a symbol is generated from a schematic's pin list (key `a`), and only the Spectre netlister reads it, which is also the one that drives VACASK. The SPICE netlister used for Ngspice ignores it. Do not delete it: without it, instances of that symbol are silently dropped from a Spectre/VACASK netlist and its `subckt` line comes out with an empty port list, with no warning at all.

#### Tip

To run a post-layout simulation, open one of the Xschem testbenches, swap the original symbol with the PEX symbol, and include the path to the SPICE netlist so the extracted netlist is used in the same testbench. This is the recommended way to compare schematic-only and post-layout simulation results.

#### Note

Note that **KLayout-PEX (KPEX)** is currently under development and therefore the Magic engine is used internally for the actual extraction.

### 5.7.4 Run Full Verification Flow at Once

If you want to run the full verification flow at once, the verify targets bundle DRC, LVS, and PEX in one command.

```
make klayout-verify [CELL=<cellname>]    # default CELL=inverter_top
make magic-verify [CELL=<cellname>]      # default CELL=inverter_top
```

## 5.8 Build Deliverables for Top-Level Integration

For the inverter to be picked up by the top-level LibreLane flow, it must expose a **LEF** (abstract pin / blockage view), a **Liberty timing library** stub, and a **Verilog stub**. The Verilog stub is referenced by `rtl/chip_top.sv` for top-level integration. These files can be built with individual Makefile targets or via `build-top`.

```
make lef          # final/lef/inverter_top.lef
make lib          # final/lib/inverter_top.lib
make verilog      # final/vh/inverter_top.vh
make build-top    # the three steps above plus copy-gds and render-gds
```

`make build-top` also copies the GDS to `final/gds/` and renders the macro to `render/img/inverter_top_white.png` and `render/img/inverter_top_black.png`, as displayed in Figure 21.

## 5.9 Full Analog Macro Flow

The full verification, build, and simulation chain is run with the following command.

```
make all
```

This calls `klayout-verify-all`, `magic-verify-all`, `build-top`, and `sim-all` (Xschem + CACE) in sequence. The simulations run last, so the `inverter_top` testbench includes the PEX netlist produced by this run. Depending on the hardware used, a full pass can take a few minutes, especially the CACE simulation step.

## 6 Bondpad and Logo Implementation

The [ip/](#) folder holds the three custom IP cells that the top-level chip relies on: a bondpad and two logos. These IPs are generated and require zero hand drawing. Each IP ships with a Python script and Makefile targets that produce the final GDS and LEF, and run KLayout and Magic DRC. This makes it easy to re-spin a different bondpad size or shape, swap the logo image, or move the artwork to another metal layer without touching layout tools.

The three IP cells covered in this section are:

- [sg13g2\\_ip\\_\\_bondpad\\_70x70](#), a  $70\text{ }\mu\text{m} \times 70\text{ }\mu\text{m}$  square bondpad built from IHP's SG13\_dev KLayout PCell library,
- [sg13g2\\_ip\\_\\_jku](#), the [Johannes Kepler University Linz](#) logo, generated from a PNG image,
- [sg13g2\\_ip\\_\\_jku\\_names](#), a companion text block with the names of the institute, generated the same way as the JKU logo.

All three IPs follow the same folder structure (`script/`, `final/`, `verification/`) and the same Makefile conventions as the digital and analog macros (`make help`, `make all`, `make clean`, `make klayout-drc`, `make magic-drc`). The logo folders `sg13g2_ip__jku` and `sg13g2_ip__jku_names` have an additional `logo/` folder that contains the source PNG image.

### 6.1 Directory Structure

The IP design files are located in [ip/](#), and the relevant folders are:

| Folder                                                                                  | Content                                                                                                                                     |
|-----------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------|
| <a href="#">sg13g2_ip__bondpad_70x70/script/</a>                                        | KLayout Python script ( <a href="#">bondpad.py</a> ) that instantiates the <code>SG13_dev::bondpad</code> PCell and writes the GDS and LEF. |
| <a href="#">sg13g2_ip__jku/logo/</a> and <a href="#">sg13g2_ip__jku_names/logo/</a>     | Source PNG images ( <a href="#">jku_logo.png</a> , <a href="#">jku_names.png</a> ) for the two logos.                                       |
| <a href="#">sg13g2_ip__jku/script/</a> and <a href="#">sg13g2_ip__jku_names/script/</a> | PNG-to-GDS converter ( <a href="#">img2lay.py</a> ) shared by both logos.                                                                   |

| Folder                                                     | Content                                                                                      |
|------------------------------------------------------------|----------------------------------------------------------------------------------------------|
| <code>final/gds/, final/lef/, final/lib/, final/vh/</code> | Build deliverables for top-level integration: GDS, LEF, Liberty, and Verilog blackbox stubs. |
| <code>verification/drc/</code>                             | DRC reports produced by <code>make klayout-drc</code> and <code>make magic-drc</code> .      |

### Tip

Read the [bondpad README](#), the [JKU logo README](#), and the [JKU names README](#) first to get familiar with the Makefile targets and parameters. Most of the commands below are also documented there.

## 6.2 Bondpad Generation

The bondpad is generated by the Python script [script/bondpad.py](#), which runs in KLayout batch mode (`klayout -zz`). It

1. instantiates the `bondpad` PCell from IHP's `SG13_dev` library,
2. flattens the PCell into a static top cell,
3. writes the layout as a GDSII file, and
4. generates a matching LEF macro description for OpenROAD / LibreLane.

Geometry and metal stack are parameterised through three Makefile variables. The defaults match the chip's padframe and the SG13G2 PDK defaults.

| Vari-<br>able             | De-<br>fault        | Description                                                                                                                                                                                                                                     |
|---------------------------|---------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <code>DIAMETER</code>     | <code>70.0</code>   | Bondpad diameter / side length in $\mu\text{m}$ .                                                                                                                                                                                               |
| <code>SHAPE</code>        | <code>square</code> | Pad geometry. Allowed: <code>square</code> , <code>octagon</code> , <code>circle</code> .                                                                                                                                                       |
| <code>BOTTOM_METAL</code> |                     | Lowest metal in the pad stack: 1= <code>Metal1</code> .. 5= <code>Metal5</code> , 6= <code>TopMetal1</code> . Top is fixed to <code>TopMetal2</code> . Default 3 matches <code>bondpad_bottomMetal</code> in <a href="#">sg13g2_tech.json</a> . |

The available Makefile targets for the bondpad are summarised below.



---

|         |                                                                                                                                            |
|---------|--------------------------------------------------------------------------------------------------------------------------------------------|
| Tar-    | What it does                                                                                                                               |
| get     |                                                                                                                                            |
| bondpad | Generate GDS and LEF via <a href="#">bondpad.py</a> (make bondpad [DIAMETER=<um>] [SHAPE=<square\ octagon\ circle>] [BOTTOM_METAL=<1-6>]). |
| verilog | Generate a Verilog blackbox stub.                                                                                                          |
| klayout | Run KLayout DRC (make klayout-drc [CELL=<cellname>] [DRC_LEVEL=<precheck\ macro\ regular>]).                                               |
| drc     | Run Magic DRC (make magic-drc [CELL=<cellname>]).                                                                                          |
| open    | Browse this folder with <a href="#">sak-open.py</a> and open each file in its tool (needs the VNC/X11 desktop).                            |
| clean   | Remove the <code>final/</code> and <code>verification/drc/</code> output directories.                                                      |

---

To execute the `clean`, `bondpad`, `verilog`, `klayout-drc`, and `magic-drc` targets in sequence, run:

```
cd ip/sg13g2_ip__bondpad_70x70
make all
```

View the bondpad layout in KLayout with:

```
ke final/gds/sg13g2_ip__bondpad_70x70.gds
```

The bondpad goes from `Metal3` up to `TopMetal2` by default, but the `BOTTOM_METAL` Makefile variable can be set to change the lowest metal in the stack. The pad is surrounded by the pad recognition layer `dfpad` (41/0).

The `bondpad` target accepts the geometry and metal-stack variables as command-line overrides. For example, to build an octagonal 80  $\mu\text{m}$  bondpad from `Metal2` up to `TopMetal2`, run:

```
make bondpad DIAMETER=80 SHAPE=octagon BOTTOM_METAL=2
```

### 6.3 Logo Generation

Both logo IPs share the same Makefile targets and PNG-to-GDS converter [script/img2lay.py](#), built on the `klayout.db` Python module. The conversion proceeds in five steps:

1. Open the source PNG (`logo/jku_logo.png` or `logo/jku_names.png`) and convert it to a binary mask using a brightness threshold (default 128).
2. Optionally invert the mask (`--invert`) so that the dark logo regions become the foreground.



Figure 25: sg13g2\_ip\_\_bondpad\_70x70 layout in KLayout.

3. For each foreground pixel, draw a square of side `--pixel-size`  $\mu\text{m}$  on the selected metal layer.
4. Merge adjacent rectangles into larger polygons (`--merge`) to keep the GDS small and DRC-friendly.
5. Add the block-bounding rectangle on the boundary layers `prBoundary` (189/0) and `NoMetFiller` (160/0).

The Makefile derives the foreground GDS layer number (e.g. 67/0 for `Metal5`) and the LEF obstruction layer name from the same `LAYER` variable, so the artwork in the GDS and the blockage exposed to the router always stay consistent.

Both logos share the same set of Makefile variables as shown below. The `jku_names` logo uses a finer standard pixel size to ensure the readability of the smaller text elements.

| Variable                 | Default (jku / jku_names)                 | Description                                                                                                                     |
|--------------------------|-------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------|
| <code>IMG_SIZE_PX</code> | 750                                       | Source image side length in pixels. Used to compute the scale factor automatically.                                             |
| <code>BLOCK_SIZE</code>  | 100                                       | Physical side length of the rendered logo in $\mu\text{m}$ .                                                                    |
| <code>PIXEL_SIZE</code>  | 0.50 / 0.25                               | Edge of one rendered pixel in $\mu\text{m}$ . Must stay the layer's minimum width (0.21 $\mu\text{m}$ on <code>Metal5</code> ). |
| <code>LAYER</code>       | <code>Metal5</code> / <code>Metal5</code> | Foreground metal layer. One of <code>Metal1</code> .. <code>Metal5</code> , <code>TopMetal1</code> , <code>TopMetal2</code> .   |

The image scale factor is computed automatically as `BLOCK_SIZE / (IMG_SIZE_PX * PIXEL_SIZE)`, so changing any of the three keeps the rendered block at the target physical size.

The available Makefile targets for the logos are summarised below.

| Target                   | What it does                                                                                                 |
|--------------------------|--------------------------------------------------------------------------------------------------------------|
| <code>logo</code>        | Convert the PNG into a GDSII layout using <a href="#">img2lay.py</a> .                                       |
| <code>lef</code>         | Emit a minimal LEF macro ( <code>CLASS BLOCK</code> , full-area OBS on <code>LAYER_NAME</code> ).            |
| <code>lib</code>         | Emit a Liberty timing stub (empty cell) so the LibreLane flow can pick the cell up.                          |
| <code>verilog</code>     | Emit a Verilog blackbox stub ( <code>module sg13g2_ip__jku;</code> ) with no ports.                          |
| <code>klayout-drc</code> | Run KLayout DRC (same <code>DRC_LEVEL</code> selection as the bondpad).                                      |
| <code>magic-drc</code>   | Run Magic DRC.                                                                                               |
| <code>open</code>        | Browse this folder with <code>sak-open.py</code> and open each file in its tool (needs the VNC/X11 desktop). |

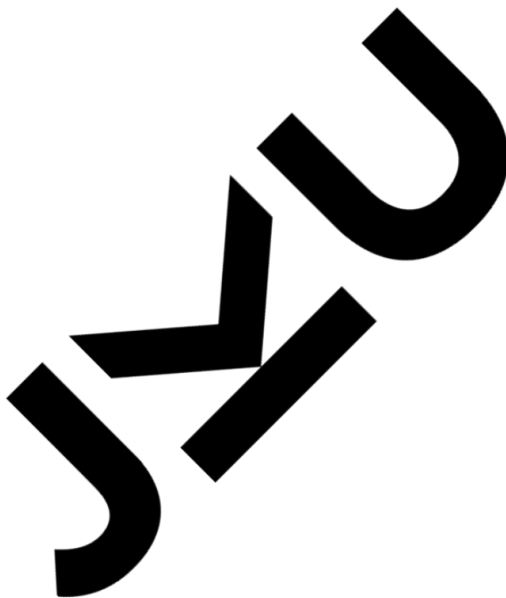
| Target             | What it does                                                    |
|--------------------|-----------------------------------------------------------------|
| <code>clean</code> | Remove <code>final/</code> and <code>verification/drc/</code> . |

To execute the `clean`, `logo`, `lef`, `lib`, `verilog`, `klayout-drc`, and `magic-drc` targets in sequence, run:

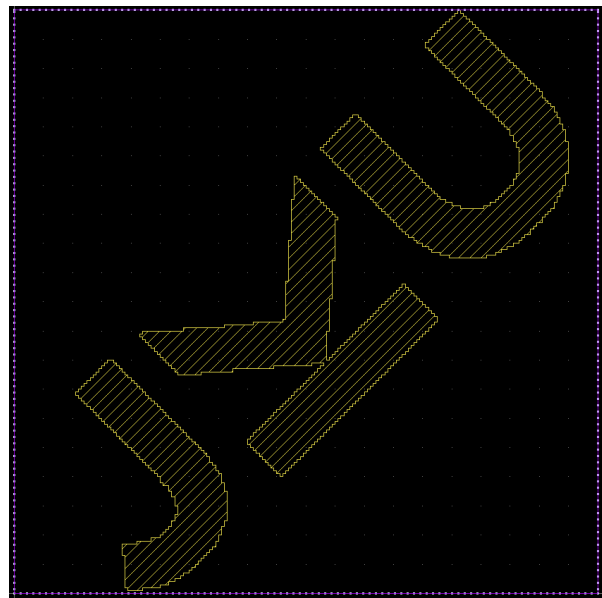
```
cd ip/sg13g2_ip__jku          # or ip/sg13g2_ip__jku_names
make all
```

View the JKU logo or JKU names layout in KLayout with:

```
ke final/gds/sg13g2_ip__jku.gds  # or sg13g2_ip__jku_names.gds
```



(a) `sg13g2_ip__jku` logo.



(b) `sg13g2_ip__jku` layout in KLayout.

Figure 26: `sg13g2_ip__jku` logo (left) and `sg13g2_ip__jku` layout in KLayout (right).

#### 💡 Tip

To replace the logo with your own artwork, drop a square PNG image into `logo/` and either rename it to `jku_logo.png` / `jku_names.png`, or override `IMG_FILE` in the Makefile. A clean black-on-white image with a few hundred pixels per side and high contrast works best.

## 6.4 Top-Level Assembly

For top-level assembly, the [top-level Makefile](#) exposes two wrapper targets that descend into the IP folders and call `make all`:

```
make build-bondpad      # build ip/sg13g2_ip__bondpad_70x70
make build-logos        # build ip/sg13g2_ip__jku and ip/sg13g2_ip__jku_names
```

Both targets are part of `make build-all` (see Section 7), so the IP cells are refreshed automatically before the chip is hardened with LibreLane. The generated GDS, LEF, Verilog stub and Liberty files are referenced by [flow/librelane/config.yaml](#) when assembling the chip top-level. Furthermore, the Verilog stubs are instantiated in [rtl/chip\\_top.sv](#).

## 7 Top-Level Implementation

This section covers the top-level assembly and padframe generation. The top-level files that need to be understood and modified for a new chip are:

- [rtl/chip\\_top.sv](#), the padframe wrapper. It instantiates one I/O cell per pad and exposes the global `clk_PAD`, `rst_n_PAD`, `input_PAD`, `output_PAD`, `bidir_PAD`, and `analog_PAD` signals.
- [rtl/chip\\_core.sv](#), the core logic. It instantiates the digital and analog macros (`counter1`, `counter2`, `inverter1`, `inverter2`, `sram_0`) and wires them together and to the I/O pads.
- [flow/librelane/config.yaml](#), the top-level LibreLane configuration. It defines the padframe order (`PAD_WEST`, `PAD_NORTH`, `PAD_SOUTH`, `PAD_EAST`), the chip area (`DIE_AREA`, `CORE_AREA`), the PDN core ring (`PDN_CORE_RING_VWIDTH`, `PDN_CORE_RING_HWIDTH`, `PDN_CORE_RING_VSPACING` and `PDN_CORE_RING_HSPACING`), the clock settings (`CLOCK_PORT`, `CLOCK_NET` and `CLOCK_PERIOD`), the macro list (`MACROS`), the bondpads, the logos, non-default routing rules (`NON_DEFAULT_RULES`, `DRT_ASSIGN_NDR`, and `RSZ_DONT_TOUCH_LIST`), STA corners, and references to `pdn_cfg.tcl` and `chip_top.sdc`.
- [flow/librelane/pdn\\_cfg.tcl](#), the Tcl script that builds the OpenROAD PDN. The bottom of the file contains the custom PDN connects for `inverter1`, `inverter2`, and `sram_0`. This is the part that has to be touched when adding new macros.
- [flow/librelane/chip\\_top.sdc](#), timing constraints (clock period, input/output delays).
- Reference documentation for the chip, [specifications.md](#), [pinout.md](#), and [floorplan.md](#).

## 7.1 Understanding `chip_top.sv`

The padframe wrapper `chip_top.sv` declares the chip's parameters and ports. The parameter block at the top of the module controls the number of pads of each type and is the place to update when the pinout changes.

The remaining code does not need to be modified for a pinout change, since the `generate` blocks automatically instantiate the correct number of I/O cells based on the parameter values. One `generate` block per pad type (`g_vdd_pads`, `g_vss_pads`, `g_iovdd_pads`, `g_iovss_pads`, `g_inputs`, `g_outputs`, `g_bidirs`, `g_analogs`) instantiates the matching IHP I/O cell, plus single instances for the clock and reset pads. An example code snippet of the input pad generation with a `generate` block is shown below. The `input_PAD` signals are exposed as ports of the `chip_top` module and are connected to the pads of the I/O cell, while the `input_PAD2CORE` signals are connected to the internal core logic in `chip_core.sv`. The same structure, or a very similar one, applies to the other pad types.

### **i** Note

Normally, only the parameter counts need to be edited in `chip_top.sv`. Only multi-clock and multi-reset chips would require modifications to the clock and reset pad generation, since the current template supports only one clock and one reset pad.

### **i** Note

The `sg13g2_IOPadAnalog` cell exposes two internal core-side nodes per analog pad:

- `padres`: path through the pad's **secondary** ESD resistor
- `padbare`: **direct** path to the pad metal (only primary ESD diodes)

In this template, `inverter2` uses both paths:

- `vin1/vin2` are connected to `analog_PADRES[0:1]` (resistor-protected input path)
- `vout1/vout2` are driven onto `analog_PADBARE[2:3]` (direct output path to avoid extra RC)

Inside `chip_core.sv`, these nets appear as the corresponding lowercase core signals `analog_padres[*]` and `analog_padbare[*]`.

---

**Listing 5** rtl/chip\_top.sv (code snippet)

---

```
generate
  for (genvar i = 0; i < NUM_ANALOG_PADS; i++) begin : g_analogs
    (* keep *)
    sg13g2_IOPadAnalog analog_pad (
      `ifdef USE_POWER_PINS
        .iovdd    (IOVDD),
        .iovss    (IOVSS),
        .vdd      (VDD),
        .vss      (VSS),
      `endif
        .pad      (analog_PAD[i]),
        .padres    (analog_PADRES[i]),
        .padbare   (analog_PADBARE[i])
    );
  end
endgenerate
```

---

## 7.2 Understanding chip\_core.sv

The core [chip\\_core.sv](#) is where the macros are instantiated and wired to the chip's I/O pads. The `enable` signal is mapped to `input_PAD[0]`, the two `counter_top` instances share the clock and reset, the two `inverter_top` instances are wired up (one digital, one analog), and the SRAM output is XOR-reduced to a single bit on `output_PAD[16]`. The input and output assignments at the top and bottom of the file correspond to the bit-to-role mapping documented in [pinout.md](#). For improved clarity, every assignment is written on a separate line. Advanced designers can also use bus notation for more compact code.

## 7.3 Understanding config.yaml

The top-level LibreLane configuration [config.yaml](#) covers the following sections and references `pdn_cfg.tcl` and `chip_top.sdc`.

1. **Padframe generation.** `PAD_WEST`, `PAD_NORTH`, `PAD_SOUTH`, and `PAD_EAST`, in the order documented in [doc/pinout.md](#).
2. **PDN core ring.** `PDN_CORE_RING`, `PDN_CORE_RING_VWIDTH`, `PDN_CORE_RING_HWIDTH`, `PDN_CORE_RING_VSPACING`, and `PDN_CORE_RING_HSPACING`.
3. **Clock settings.** `CLOCK_PORT`, `CLOCK_NET`, and `CLOCK_PERIOD`.

#### 4. STA corners.

- `nom_fast_1p32V_m40C`
- `nom_fast_1p65V_m40C`
- `nom_slow_1p08V_125C`
- `nom_slow_1p35V_125C`
- `nom_typ_1p20V_25C`
- `nom_typ_1p50V_25C`

#### 5. Non-default routing rules. `NON_DEFAULT_RULES`, `DRT_ASSIGN_NDR`, and `RSZ_DONT_TOUCH_LIST`.

#### 6. Macro, bondpad, and logo placement. `MACROS` with per-macro files and placement (location, orientation), plus top-level bondpad and logo placement settings, together with chip area and placement constraints (`DIE_AREA`, `CORE_AREA`, `FP_SIZING`, `PL_TARGET_DENSITY_PCT`).

#### Note

The `NON_DEFAULT_RULES` section defines a dedicated routing rule for the analog traces of `inverter2` (`PADRES` and `PADBARE`). In this template, `NDR_analog` sets a wider wire width and larger spacing for the selected metals and vias, which helps reduce coupling and preserve analog signal quality.

The `DRT_ASSIGN_NDR` section applies this rule during detailed routing by matching net names with regular expressions (`^.*PADRES.*$` and `^.*PADBARE.*$`). The `RSZ_DONT_TOUCH_LIST` then protects the same analog nets from resizing, buffering, or other optimization steps, so the routed analog paths remain unchanged.

## 7.4 Understanding `pdn_cfg.tcl`

`pdn_cfg.tcl` is the OpenROAD callback that builds the power-distribution network (PDN). Most of the file follows the standard structure provided by LibreLane and OpenLane.

For macro changes, the bottom of the file is the relevant part. It defines per-instance PDN grids and macro connections. The current chip provides the following.

- A default macro grid (`TopMetal2` to `TopMetal1`) used by the two counters.
- A custom grid for `inverter1` and `inverter2` (`TopMetal2` to `TopMetal1` and `TopMetal2` to `Metal5`, because each inverter macro has its own internal `Metal5` / `TopMetal1` ring).
- A custom grid for `sram_0` (`Metal5` stripes bridged to `Metal4` and `TopMetal1`, because the SRAM exposes its supplies on `Metal4` and `Metal5`).

When new macros are added, the matching `define_pdn_grid` plus `add_pdn_connect` block at the bottom must be extended (or the default grid is used if the macro layers line up with the



top-level grid). The custom grid for the inverters is shown below as an example. The comments explain the reasoning behind the layer choices and the connection structure.

## 7.5 Top-Level Simulation

The same RTL and gate-level cocotb flow as in Section 4 is available at top level.

```
make sim-rtl-cocotb
make sim-gl-cocotb
make sim-view-cocotb # use GTKWave as waveform viewer
make sim-view-cocotb WAVEFORM_VIEWER=surfer # use Surfer as waveform viewer
```

The cocotb testbench at top level is [testbenches/cocotb/chip\\_top\\_tb.py](#), and it covers the global enable, reset, counters, and bidir behaviour documented in [doc/specifications.md](#). The result is shown in Figure 27.

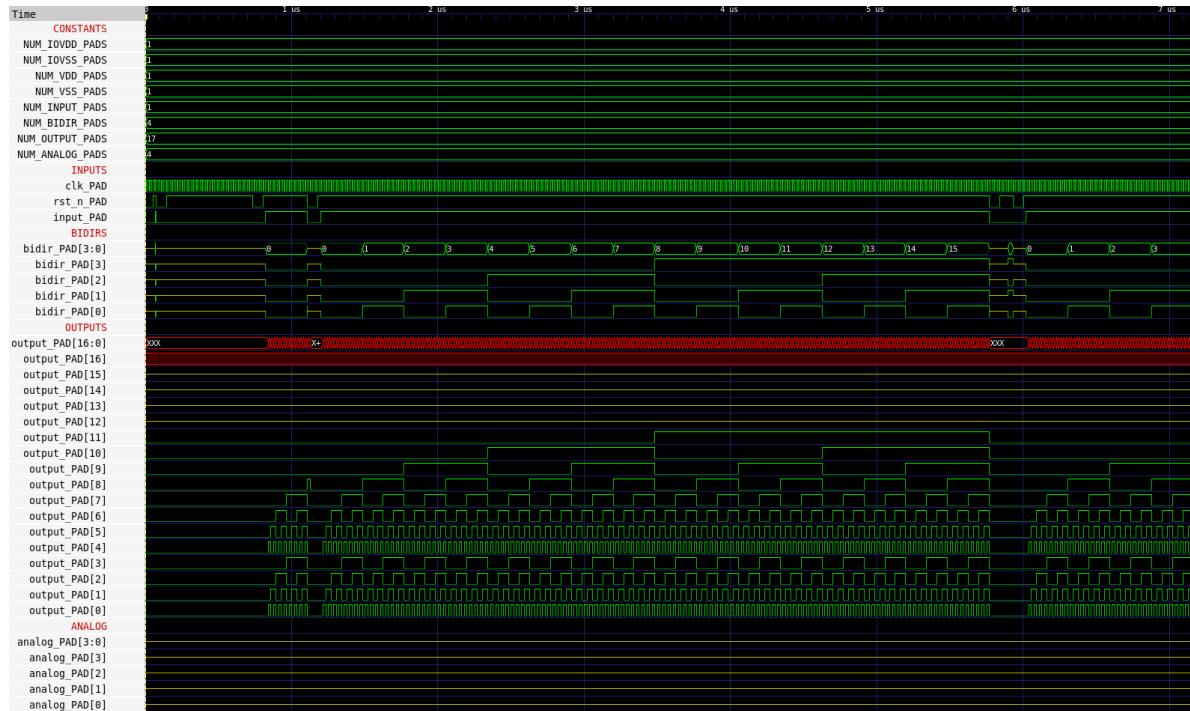


Figure 27: chip\_top cocotb waveform in GTKWave.

For mixed-signal simulation, the [testbenches/xschem/chip\\_top\\_tb\\_tran.sch](#) testbench can be opened in Xschem for interactive simulation, the same way as for the counter or the inverter, and is shown in Figure 28.

```
cd testbenches/xschem
xschem chip_top_tb_tran.sch
```

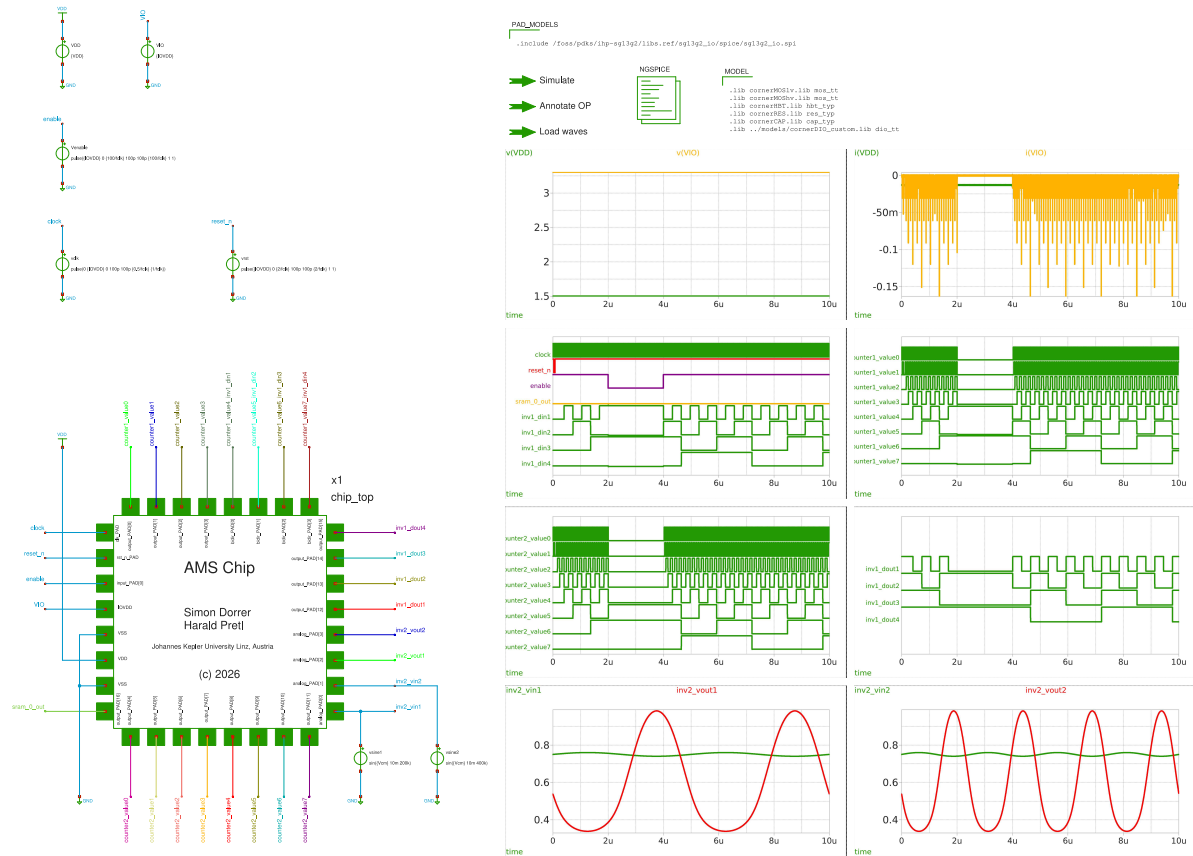


Figure 28: Xschem top-level mixed-signal testbench `chip_top_tb_tran`.

Again, the simulation and plotting can be run non-interactively from the command line with the following targets.

```
make sim-gl-xschem
make sim-view-xschem SCRIPT=plot_chip_top
```

`sim-gl-xschem` runs in batch mode: it netlists the testbench with `xschem netlist` and then calls `ngspice -b` directly, so `make` blocks until the run finishes and sees its exit status. The testbench is selected with `TB=<testbench_name>` (default: `<CELL>_tb_tran`). Since the run is headless, the `plot` commands in the testbench's `.control` block are a no-op. The testbench exports its results with `wrdata` to `testbenches/xschem/plot_simulations/data/`, from where `sim-view-xschem` plots them with Matplotlib.

### **i** Note

The top-level Xschem testbench converges, but it may take a long time depending on the hardware used. `make sim-gl-xschem` is part of `sim-all`, so a full `make sim-all` includes it. For this tutorial, the simulation data is already available in the repository in `testbenches/xschem/plot_simulations/data/chip_top_tb_tran.txt`, so `make sim-view-xschem SCRIPT=plot_chip_top` can be run without running the simulation first.

The following plots show the results of the top-level transient simulation. The first plot shows the analog inverter response, while the other three plots show the digital outputs of the two counters, the digital inverter, and the bidirs.

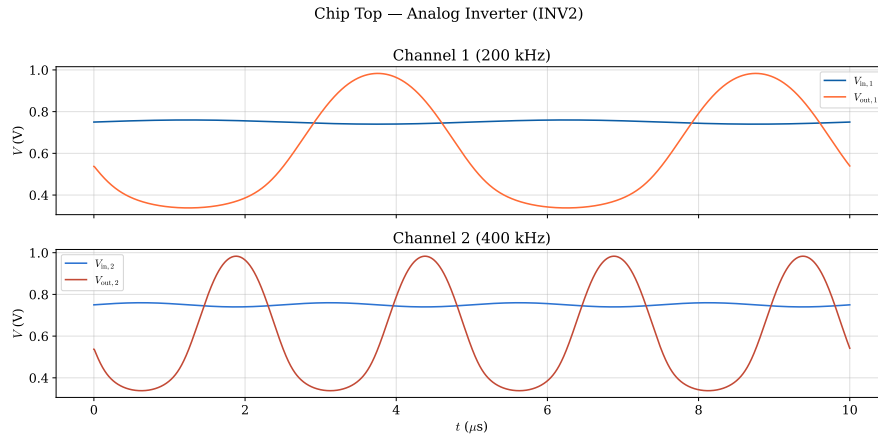


Figure 29: Plotted simulation result of `chip_top_tb_tran`: analog inverter 2 response.

When the testbench is opened in Xschem, the schematic of `chip_top` can be inspected with the shortcuts `e` / `Ctrl + e`, and the symbol can be shown with `i` / `Ctrl + e` as explained in Section 5. The symbol and schematic are shown in Figure 33 and Figure 34, respectively. The schematic shows the connections between the padframe and the core macros, while the symbol shows the pinout of the chip. The SRAM is not included in the schematic, since IHP does not ship a SPICE model or an Xschem symbol for the `RM_IHPSG13_1P_1024x32_c2_bm_bist` macro.

## 7.6 Building the Chip

When the floorplan and pinout are planned and configured, the chip can be built from scratch with the following command.

```
make build-all
```

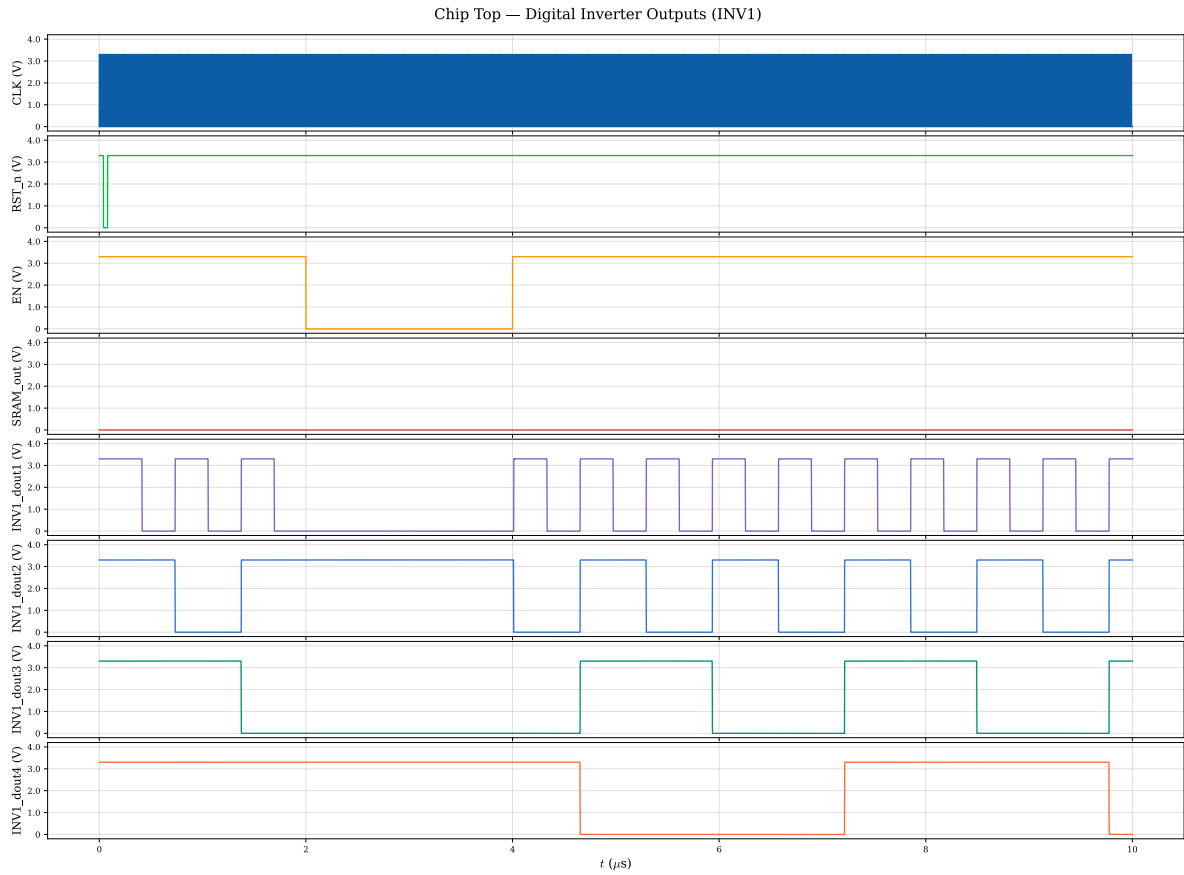


Figure 30: Plotted simulation result of chip\_top\_tb\_tran: digital inverter 1 outputs.

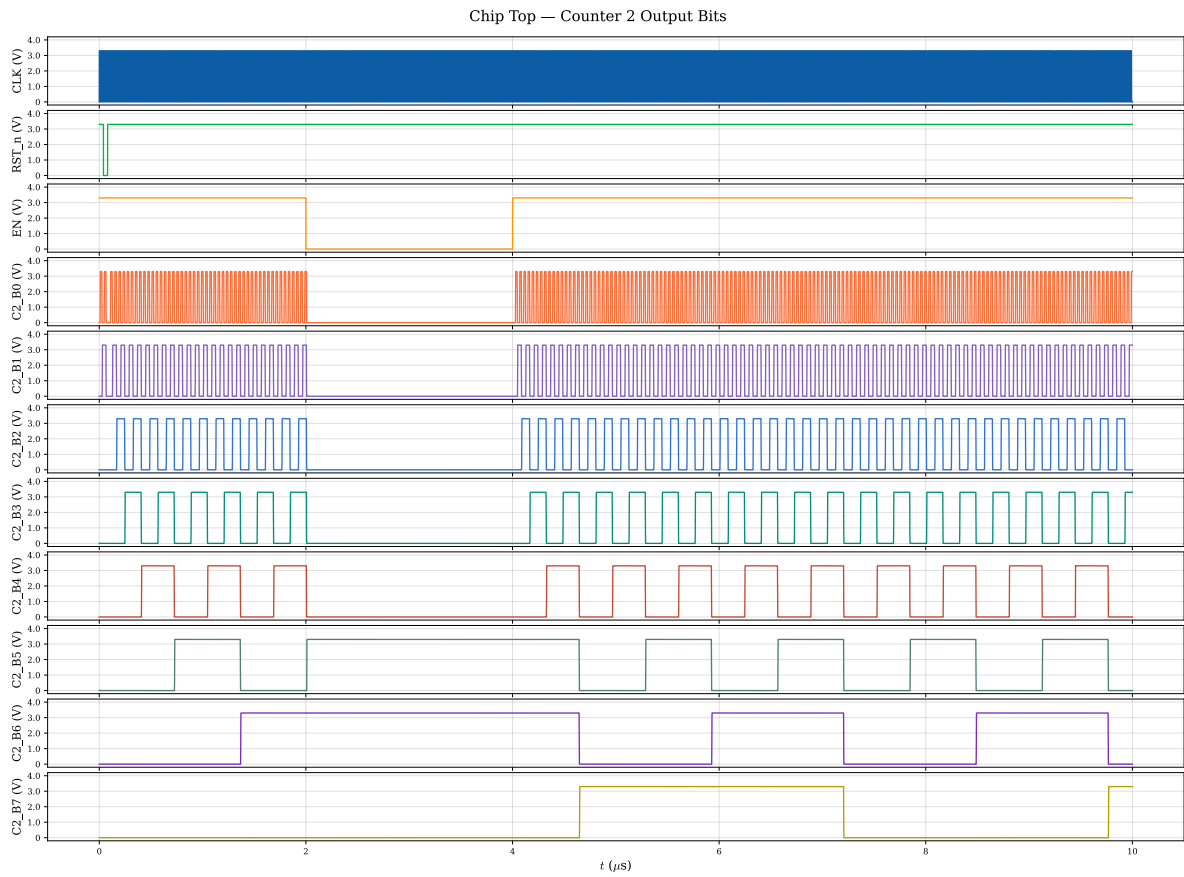


Figure 31: Plotted simulation result of `chip_top_tb_tran`: digital counter 2 outputs.

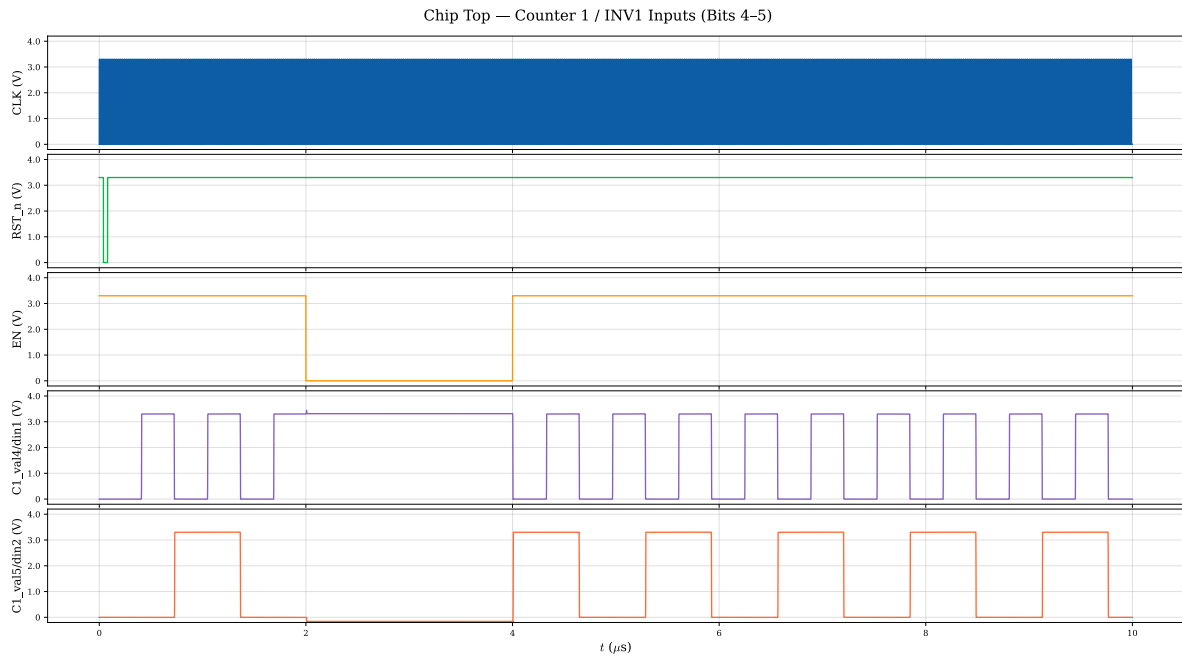


Figure 32: Plotted simulation result of `chip_top_tb_tran`: digital counter 1 / inverter 1 bidirs.

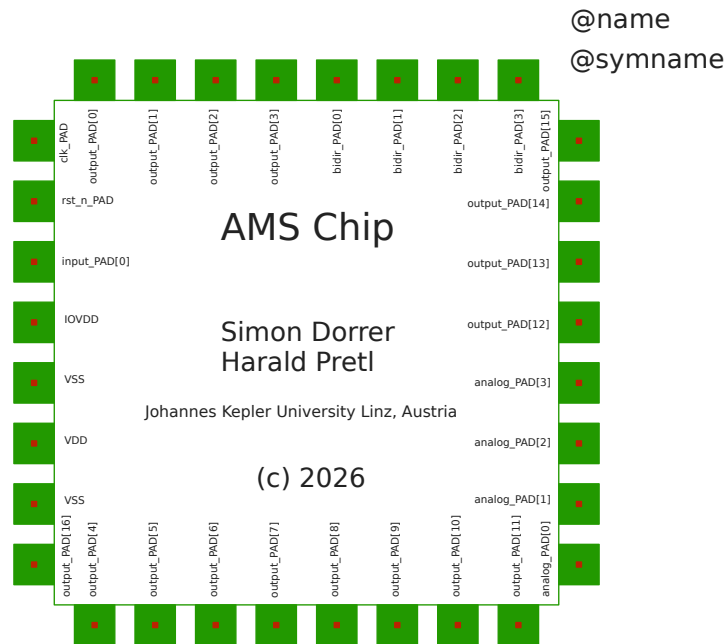


Figure 33: Xschem symbol of `chip_top`, showing the pinout of the chip.

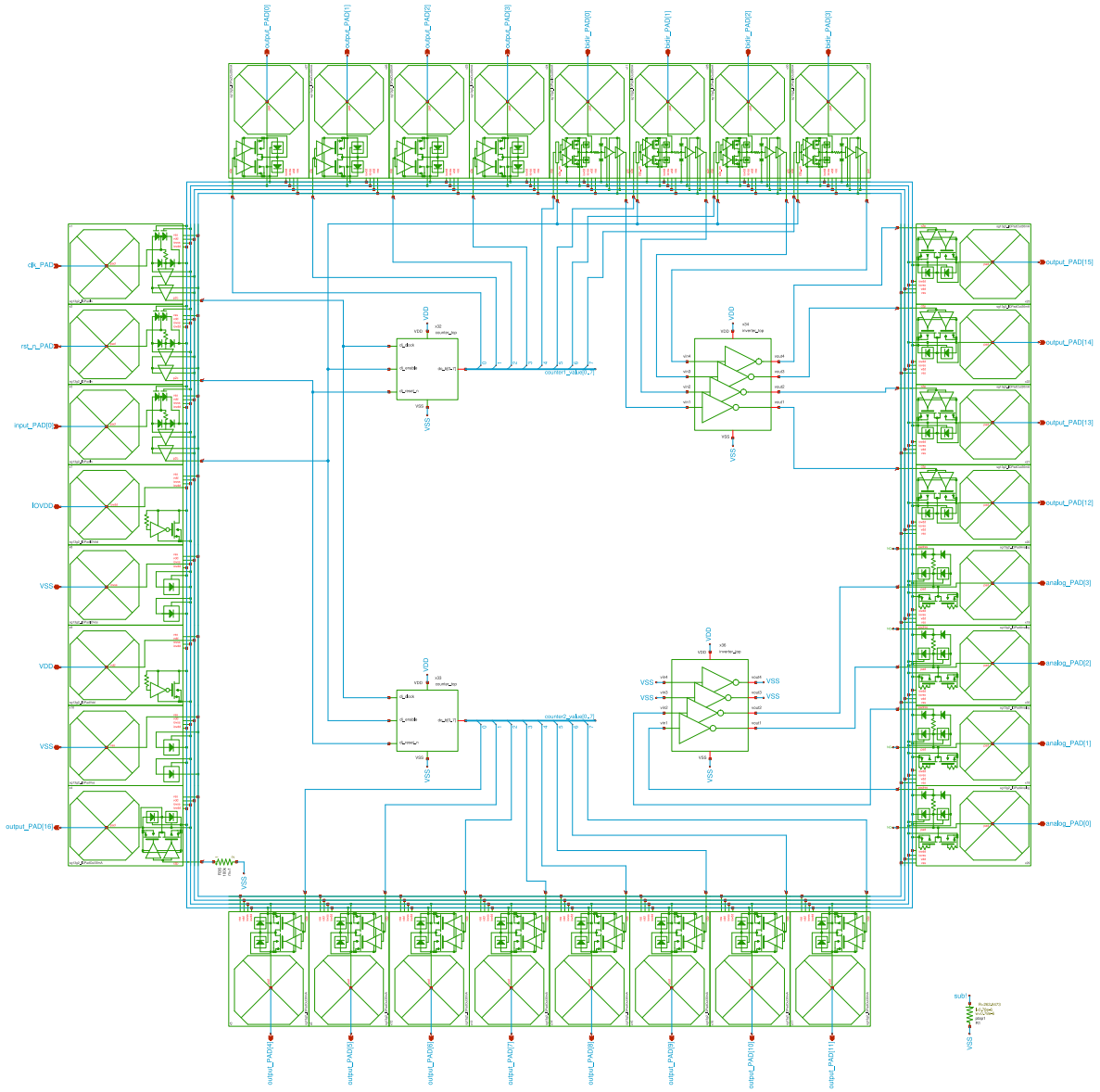


Figure 34: Xschem schematic of `chip_top`, showing the connections between the padframe and the core macros.

This runs the steps in order, `init-submodules`, `build-bondpad`, `build-logos`, `build-macros`, and `build-top`. It is the right starting point on a fresh machine after cloning the repository and entering the IIC-OSIC-TOOLS container.

To re-run only the top-level flow (for example after a change in `config.yaml` or `chip_core.sv`), without rebuilding the bondpad, logos, or macros, run the following command.

```
make build-top
```

Internally, `build-top` runs `librelane-nodrc`, `copy-reports`, `copy-gds`, `copy-netlist`, `copy-render`, `add-logo-fill`, and `render-gds`. The final GDS is written to `layout/chip_top_logo_fill.gds`.

The renders generated by `sak-render.py`, which runs during the `render-gds` step, are located in `render/img/`. The verification reports (Yosys, antenna, STA, IR-drop, LVS, manufacturability) from the top-level LibreLane flow can be found in `verification/reports/`. The netlists are located in `netlist/`.

#### Note

Note that currently the LibreLane flow streams out the top-level GDS `layout/chip_top.gds.gz` without the logos and the fill structures. To add the chip logo (PNG → GDS) and the fill structures on top of the LibreLane output, run:

```
make add-logo-fill
```

This calls `scripts/add_logo_fill.sh` and writes `layout/chip_top_logo_fill.gds.gz`. This step is already called by `make build-top`. In the future, it is planned to replace this script and Makefile target with a custom LibreLane step.

#### Tip

The difference between `chip_top.gds.gz` and `chip_top_logo_fill.gds.gz` is that the former is the raw top-level layout without any logos or filler cells, while the latter includes the logos and filler cells added for manufacturability and cosmetic purposes. For visual debugging in KLayout, it is recommended to use `chip_top.gds.gz` to avoid the visual clutter of the logos and filler, while for DRC and LVS, `chip_top_logo_fill.gds.gz` should be used since it is the actual layout that will be taped out.

## 7.7 Viewing the Chip

After a successful build, the chip can be opened with KLayout.



```
ke layout/chip_top.gds.gz          # Chip layout without logos and filler
ke layout/chip_top_logo_fill.gds.gz # Chip layout with logos and filler
```

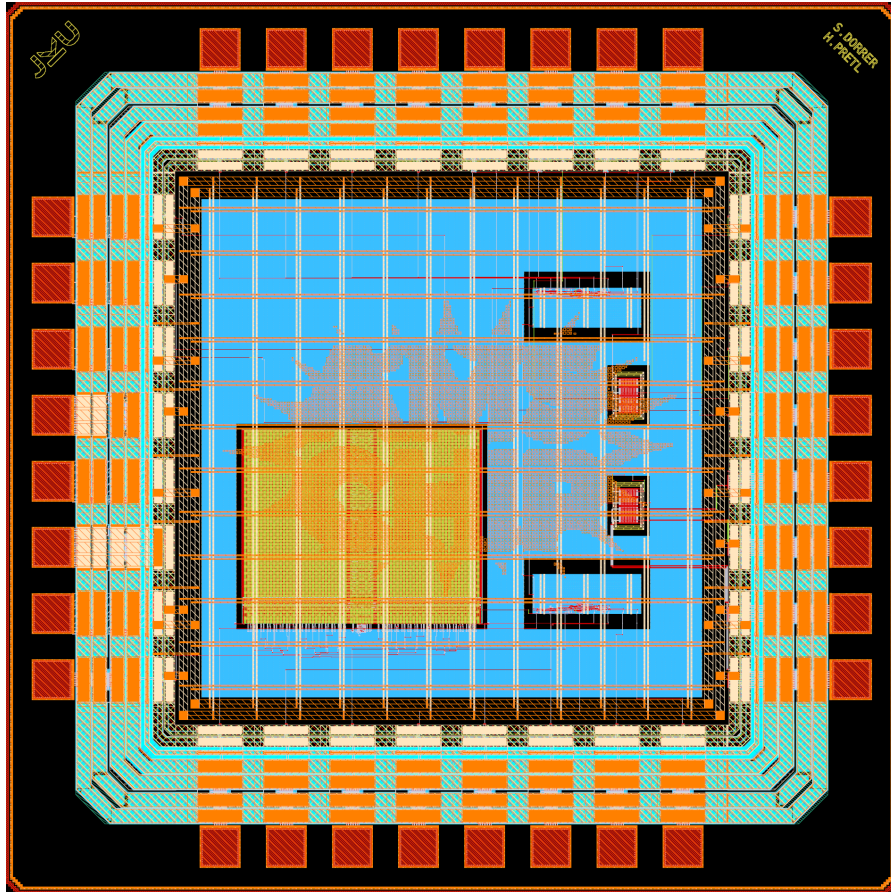


Figure 35: chip\_top\_logo\_fill layout in KLayout (“all” layers).

## 7.8 Top-Level Verification

DRC, LVS, and PEX are also available for the full chip.

### 7.8.1 DRC

The following DRC targets are available for the top-level.

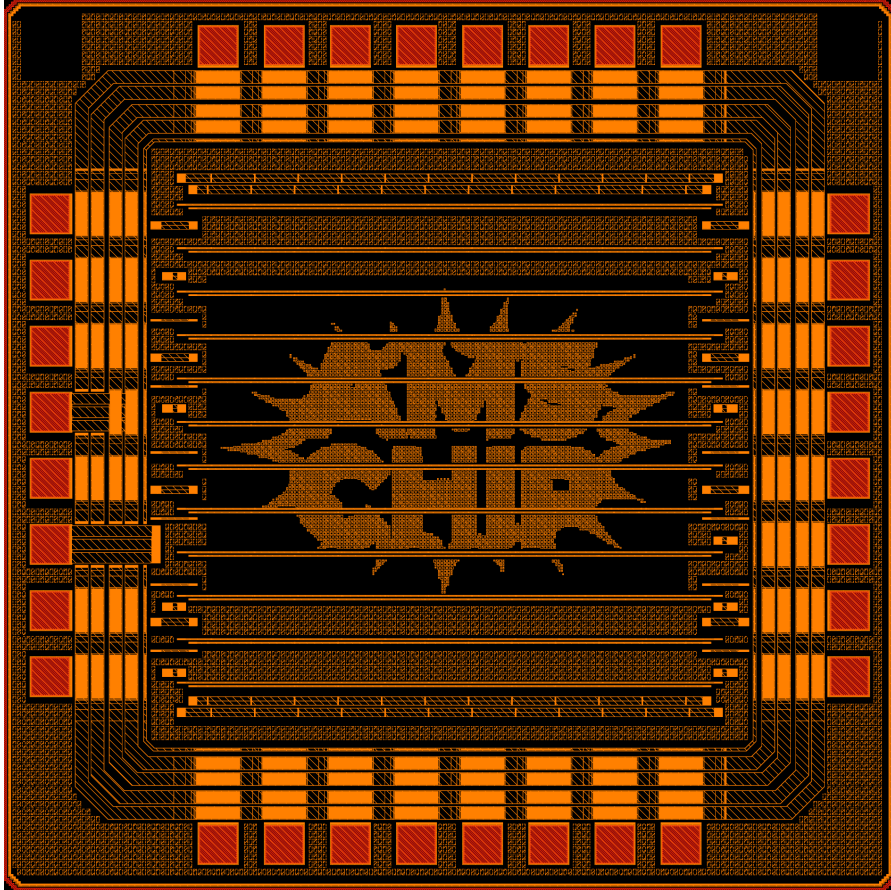


Figure 36: chip\_top\_logo\_fill layout in KLayout with only TopMetal2 selected.

```

make magic-drc CELL=chip_top           # Magic DRC on the bare chip
make magic-drc CELL=chip_top_logo_fill # Magic DRC including logo + filler
make klayout-drc-minimum                # fast pre-check KLayout DRC on chip_top_logo_fill
make klayout-drc-regular                # full KLayout DRC on chip_top_logo_fill.gds.gz (s

```

`klayout-drc-minimum` is a quick pre-check on the final top-level layout including the logo and the fill structures. `klayout-drc-regular` does the full check, but takes longer. `magic-drc` is faster than `klayout-drc`, but it does not support the full set of DRC rules and is less accurate on the complex top-level layout, so it is recommended to use both.

### 7.8.2 LVS

The following LVS targets are available for the top-level.

```

make klayout-lvs          # top-level KLayout LVS
make magic-lvs            # top-level Magic + Netgen LVS

```

#### Warning

The top-level LVS is currently under construction. The flow is in place, but some issues and limitations in the ihp-sg13g2 PDK must be fixed before it can run successfully.

### 7.8.3 PEX

The following PEX target is available for the top-level.

```

make magic-pex          # only chip_top (without logo and filler), default
make magic-pex EXT_MODE=3 THRESHOLD=5000 MINRES=500 MINDELAY=2 # full-RC with custom extre

```

For full-RC extraction (`EXT_MODE=3`), `magic-pex` also exposes the `sak-pex.sh` `extresist` tuning parameters `THRESHOLD` (`-t`, default 10000 mOhm), `MINRES` (`-r`, default 1000 mOhm) and `MINDELAY` (`-y`, default 1 ps; 0 = gate by resistance). They are ignored in `EXT_MODE=1/2`.

As in the macros, `magic-pex` starts by running `make symbol-pex`, which rebuilds `schematic/xschem/chip_top_pex.sym` from `chip_top.sym` with `type=subcircuit` changed into `type=primitive`, so the PEX symbol always matches the current chip symbol.

### Warning

At the top level, PEX is only run on `chip_top` (without logo and filler), and the default `EXT_MODE=1` (C-decoupled) is used to keep the runtime manageable. However, the extracted netlist `chip_top_magic_pex_1.spice` is still huge (~75 MB, ~566k lines), and the post-layout simulation must be run on a powerful server.

## 7.9 Packaging

The die is packaged in a QFN-32 ( $5 \times 5$  mm) open-molded plastic package ([OmPP](#)) from the [EUROPRACTICE ASIC packaging](#) offering. The package matches the padframe exactly: all 32 bondpads (8 per side, see [Section 7](#)) are wired one-to-one to the 32 package leads (8 per side). Bonding is done with 25  $\mu$ m gold wire, the longest wire measures 1.70 mm, and the cover is glued on after bonding.

The bonding diagram below is generated fully automatically with `make bondplan` from the final chip GDS ([layout/chip\\_top\\_logo\\_fill.gds.gz](#)) and the EUROPRACTICE package library [EP\\_PACKAGES\\_08022018.gds](#). The flow extracts the die bondpads together with their names, places the die in the package cavity, draws the bondwires, fills in the title block, and checks wire lengths, crossings, spacing, lead skew, and the clearance around the sensitive analog wires. The [pinout](#) and all flow settings are kept in a single configuration file, so the diagram never goes out of sync with the chip. Details are documented in the [packaging README](#).

```
make bondplan                # uses the default VERSION (1.0.0)
make bondplan VERSION=2.1.0  # stamp another version on the sheet
```

The `VERSION` variable is passed to the flow and printed in the title block (DIE: CHIP\_TOP - V.1.0.0), so the version number is maintained in the Makefile only. The outputs are the bondplan GDS (`packaging/layout/chip_top_bondplan.gds`), the bond report with summary and bond table ([packaging/result.md](#)), and the bonding diagram images (`packaging/render/chip_top_bondplan_{white,black}.{png,svg}`).

### Tip

Check the flow log and [packaging/result.md](#) after every run. Unbonded named pads and NC package pins are reported explicitly, so a forgotten connection in the `PINOUT` table is immediately visible. Warnings are raised for bondwires that are too long (`BONDWIRE_MAX_LENGTH`), cross each other, come too close to each other, land skewed on their lead (`BONDWIRE_MAX_SKEW`), or violate the guard clearance of sensitive wires (`GUARDED_PINS` / `GUARD_SPACING`).

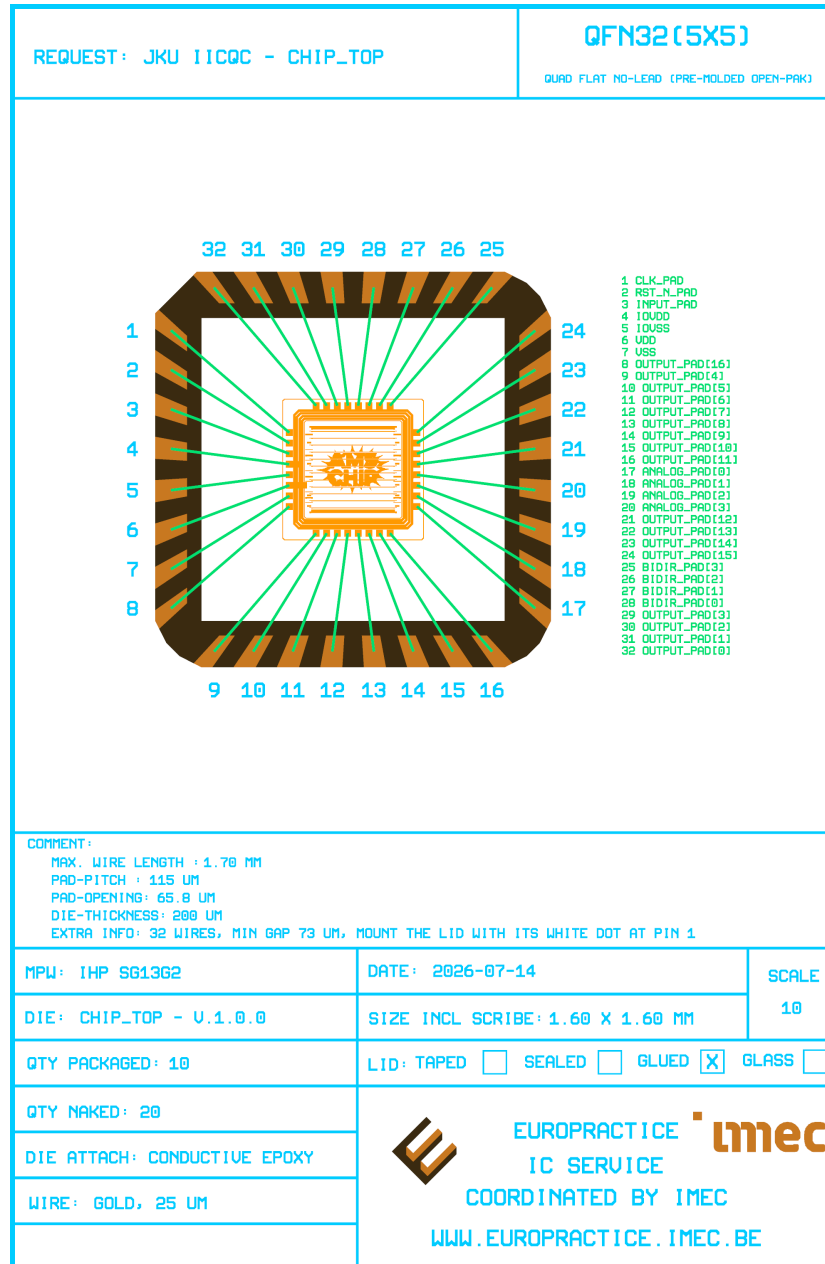


Figure 37: Automatically generated bonding diagram of the ihp-sg13g2 AMS template chip in a QFN-32 package.

## 7.10 Full Flow at Once

The comprehensive simulation, build, and DRC pass for the complete chip is run with the following command.

```
make all
```

This calls `build-all` first, then Magic DRC on both `chip_top` and `chip_top_logo_fill`, then `sim-all` (RTL and gate-level cocotb plus the gate-level Xschem run), and finally `bondplan`. The simulations therefore run on the netlists that this build has just produced. To shorten the runtime, the KLayout DRC is currently not included in `make all`. However, depending on the hardware used, `make all` can still take a while.

## 7.11 Releasing for Tapeout

When the chip is clean, the `release` target copies the GDS, the netlists, and the renders into a versioned folder for tapeout submission.

```
make release VERSION=2.1.0
```

The artifacts are written to `release/v.2.1.0/` (`gds/`, `netlist/{schematic,layout,pnl,spice}/`, `img/`).

### ! Important

Run `make all` before `make release`. The release target only copies files, it does not rebuild or re-verify the chip.

## 8 Additional Exercises

The following exercises repeat the corresponding sections of this tutorial with modifications to the design.

### 8.1 Exercise 1: From 8-bit Up-Counter to 16-bit Down-Counter

Repeat Section 4 for a 16-bit synchronous down-counter that wraps from 0 back to  $2^{16} - 1 = 65535$ . The recommended steps are:

1. Update the RTL (`counter.sv`, `counter_top.sv`, `constants.sv`). Change the bitwidth and turn the increment block into a decrement, with wrap-around from 0 to `CTR_MAX`.



2. Update the testbenches in `testbenches/verilog/`, `testbenches/cocotb/`, and `testbenches/xschem/` to cover the new behaviour. Adjust the bus width in the waveform view files (`*.gtkw`, `*.surf.ron`) as well.
3. Run `make sim-all` to validate the new RTL.
4. Re-run the LibreLane flow with `make build-top`. Check the timing reports. A 16-bit counter at 50 MHz must still pass timing constraints.
5. Update the bus range of the output pin in the Xschem symbol (`counter_top.sym`), and re-run the gate-level Xschem simulation. The XSPICE model itself is regenerated automatically by `make build-top` (or manually with `make generate-xspice`), and the PEX symbol `counter_top_pex.sym` together with the extracted netlist by `make magic-pex`. Forgetting this step no longer costs you a debugging session: `make symbol-check` runs inside `generate-xspice` and names the eight ports the symbol is now missing. Run `make symbol-gl FORCE=1` in a scratch copy of the macro if you would rather read the widened pin off a fresh scaffold than edit the range by hand.

💡 Bonus for advanced designers

Replace the binary counter with a 16-bit Gray counter (only one bit toggles per transition).

## 8.2 Exercise 2: Self-Biased Single-Ended Inverter Amplifier

Repeat Section 5 with a self-biased single-ended inverter amplifier by adding a feedback resistor  $R_f$  between the output and the input of the inverter. As discussed in Section 5.4, the bare CMOS inverter has poor PVT robustness because nothing pins the gate bias to  $V_{DD}/2$ . A feedback resistor closes a DC loop that forces the operating point to the high-gain crossover ( $V_{in} \approx V_{out} \approx V_{DD}/2$ ) regardless of process corner, which also linearises the small-signal transfer characteristic and turns the inverter into a usable single-ended amplifier. The disadvantage of this technique is the reduced input resistance, which equals  $R_{in} \approx 1/(g_{m,NMOS} + g_{m,PMOS})$ . The recommended steps are:

1. Open the inverter schematic (`inverter.sch`) in Xschem and add a resistor between the `vin` and `vout` nets of the inverter. Start with a value of  $R_f = 1\text{ M}$  and adjust as needed. What can you observe?
2. Re-run the schematic-level testbenches from Section 5 and observe the change in behaviour. First, always carefully think about what the expected behaviour would be before running the simulations, and then check if the results match your expectations.
3. Re-run the full schematic-level simulation suite with `make sim-all` to refresh the plotted results in `fig/inverter_top/`.
4. Re-run the CACE flow with `make sim-cace` and compare the new process and mismatch plots against Figure 17 and Figure 19. The spread of  $A_{ol}$  across corners should be visibly narrower than the open-loop reference.

5. [Optional] Sweep  $R_f$  in the CACE characterisation YAML ([inverter.yaml](#)) to study the circuit behaviour across a range of feedback resistor values.
6. [Optional] Once the schematic results are convincing, propagate the resistor to the layout in [layout/](#) and re-run the verification chain (`make klayout-verify`, `make magic-verify`) plus `make build-top`.

#### Tip

Choose  $R_f$  carefully. Too small, and the loop loads the amplifier output and degrades the closed-loop gain. Too large, and the time constant with the input parasitic capacitance dominates, so the amplifier loses gain at the low end of the signal band. A value in the 1–10 M range is a sensible starting point for this inverter sizing.

### 8.3 Exercise 3: Three-Stage Ring Oscillator with Output Buffer

Repeat Section 5 for a three-stage ring oscillator with an output buffer, derived from the existing quad inverter. The recommended steps are:

1. Copy the [inverter](#) macro folder to a new location (for example `macros/ringosc/`), and adapt the Makefile names.
2. Update the schematic in Xschem. Connect three inverters in a ring, and tap the output of the last stage through a buffer. The fourth inverter of the quad inverter can naturally be used as the buffer.
3. Update the symbol of the macro ([schematic/xschem/inverter\\_top.sym](#)) to expose a single output net, instead of `vin1..vin4` and `vout1..vout4`. The PEX symbol `inverter_top_pex.sym` is rebuilt from it by `make symbol-pex`, which the PEX targets run automatically.
4. Update the transient testbench ([testbenches/xschem/inverter\\_top\\_tb\\_tran.sch](#)) to measure the oscillation frequency. Delete testbenches that are no longer relevant.
5. [Optional] Adapt the CACE [inverter.yaml](#) to characterise the oscillation frequency across PVT.
6. Update the layout in KLayout. The existing `inverter_top.klay.gds` can be used as a starting point, but the internal routing has to be modified to form the ring and the buffer stage. After editing, export a clean `ringosc_top.gds`.
7. Re-run the verification chain, `make klayout-verify`, `make magic-verify`, and `make build-top`.

### 8.4 Exercise 4: Update Pinout and Floorplan of the Top-Level

Building on Section 8.1 and Section 8.3, swap `inverter2` for the ring oscillator, delete `counter2`, and connect the now 16-bit `counter1` to the freed-up south-side pads. The recommended steps



are:

1. Replace `inverter2` with the ring oscillator macro in `chip_core.sv`.
2. Replace three of the four analog pads with VSS pads, and connect the ring oscillator's output to the fourth analog pad.
3. Delete `counter2` and rewire all eight south-side pads to the new 16-bit `counter1` values.
4. Update the parameters of `chip_top.sv` so that `NUM_OUTPUT_PADS`, `NUM_BIDIR_PADS`, `NUM_ANALOG_PADS`, and `NUM_VSS_PADS` match the new pin count (see Section 7).
5. Update the macro instantiation and wiring in `chip_core.sv`.
6. Update the placement coordinates in the `MACROS` block of `config.yaml`. Keep the `doc/floorplan.md` constraints in mind, namely stay in the core, no overlaps, and respect the Metal3 grid for inverter-like macros.
7. Update the PAD lists (`PAD_WEST`, `PAD_NORTH`, `PAD_SOUTH`, `PAD_EAST`) in `config.yaml` to match the new pinout.
8. Update `pdn_cfg.tcl` so that the new macros are covered by the correct PDN grid (default for digital, `inverter_top`-style for analog).
9. Rebuild the chip from scratch with `make build-all`, then verify with `make magic-drc CELL=chip_top` and `make klayout-drc-minimum`.

## 8.5 Exercise 5: Update the Bondplan Configuration

Building on Section 8.4, update the bondplan configuration in `packaging/config.yaml` so that the packaged chip matches the new pinout (ring oscillator instead of `inverter2`, 16-bit `counter1`, no `counter2`, three additional VSS pads). The recommended steps are:

1. Rebuild the final GDS of the modified chip (`make build-all` from Section 8.4), since the bondplan flow reads the die pads and their names directly from `layout/chip_top_logo_fill.gds.gz`.
2. Update the PINOUT table in `packaging/config.yaml`: the south-side pads now carry `counter1_value[15:8]`, three of the four east-side analog pads became VSS pads, and the fourth carries the ring-oscillator output. Keep the geometric order in mind (pins 1-8 left top→bottom, 9-16 bottom left→right, 17-24 right bottom→top, 25-32 top right→left) so that the bondwires do not cross.
3. Connect the freed-up VSS pads as downbonds to the exposed pad by adding an EPAD list (e.g. `EPAD: ["VSS", "VSS", "VSS"]`) and marking their former leads as NC (~). The exposed pad of the QFN-32 is the package ground, and the die is attached with conductive epoxy, so ground downbonds are shorter and lower-inductance than wires to leads.
4. Re-run `make bondplan` and read the flow log and `packaging/result.md` carefully: only the deliberately unbonded pads may appear as *unbonded named pads* (plus the seal-ring artifact), and there must be no wire-length, crossing, spacing, lead-skew, or guard-clearance warnings.

5. Inspect the generated bonding diagram in [packaging/render/](#) and the bondplan GDS in KLayout (`ke packaging/layout/chip_top_bondplan.gds`). The title block of the A4 drawing sheet (die size, pad pitch, wire count, maximum wire length) is refilled automatically on every run.
6. [Optional] Experiment with the die placement: rotate the die with `DIE_PLACEMENT.orientation` (e.g. E) or move it off-center with `DIE_PLACEMENT.location`, re-run the flow, and observe how the wire-length, spacing, and skew checks react. Restore a placement where all checks pass.

## 9 RFIC Flow

Are you interested in an open-source RFIC flow? Check it out [here](#).

## 10 Acknowledgements

This project is funded by the JKU/SAL [IWS Lab](#), a collaboration of [Johannes Kepler University](#) and [Silicon Austria Labs](#).

---

**Listing 1** macros/counter/rtl/counter.sv

---

```
// SPDX-FileCopyrightText: 2026 Simon Dorrer and Harald Pretl
// SPDX-License-Identifier: Apache-2.0 WITH SHL-2.1
// Description: This file implements an N-bit up counter with
// synchronous reset & enable in SystemVerilog.

`default_nettype none
`ifndef __COUNTER__
`define __COUNTER__

module counter #(
    parameter int unsigned    CTR_BW  = 8,
    parameter logic [CTR_BW-1:0] CTR_MAX = 2**CTR_BW - 1
)(
    input logic                clock_i,
    input logic                reset_i,
    input logic                enable_i,

    output logic [CTR_BW-1:0] counter_value_o
);

    // Counter implementation
    always_ff @(posedge clock_i) begin
        if (reset_i) begin
            // Synchronous reset clears the counter value
            counter_value_o <= '0;
        end else if (enable_i) begin
            // Increment the counter value by 1, wrap around at CTR_MAX
            if (counter_value_o == CTR_MAX) begin
                counter_value_o <= '0;
            end else begin
                counter_value_o <= counter_value_o + 1;
            end
        end
    end

endmodule // counter

`endif
`default_nettype wire
```

---

---

**Listing 2** macros/counter/rtl/counter\_top.sv

---

```
// SPDX-FileCopyrightText: 2026 Simon Dorrer and Harald Pretl
// SPDX-License-Identifier: Apache-2.0 WITH SHL-2.1
// Description: This file implements the top-level wrapper module
// of the counter macro in SystemVerilog.

`default_nettype none
`ifndef __COUNTER_TOP__
`define __COUNTER_TOP__

module counter_top #(
    parameter int unsigned CTR_MAX = `COUNTER_MAX_DEFAULT,
    localparam int unsigned CTR_BW = $clog2(CTR_MAX + 1)
) (
    input logic          clock_i,
    input logic          reset_n_i,
    input logic          enable_i,

    output logic [CTR_BW-1:0] counter_value_o
);

    // Internal active-high reset (wrapper handles polarity conversion)
    logic reset;
    assign reset = ~reset_n_i;

    // Embed Counter
    counter #(
        .CTR_BW(CTR_BW),
        .CTR_MAX(CTR_MAX)
    ) counter_0 (
        .clock_i(clock_i),
        .reset_i(reset),
        .enable_i(enable_i),
        .counter_value_o(counter_value_o)
    );
endmodule // counter_top

`endif
`default_nettype wire
```

---

---

**Listing 3** rtl/chip\_top.sv (code snippet)

---

```
module chip_top #(
    // Power / ground pads for digital core / analog front-end
    parameter int unsigned NUM_VDD_PADS    = 1,
    parameter int unsigned NUM_VSS_PADS    = 1,

    // Power / ground pads for I/O
    parameter int unsigned NUM_IOVDD_PADS  = 1,
    parameter int unsigned NUM_IOVSS_PADS  = 1,

    // Signal pads
    parameter int unsigned NUM_INPUT_PADS  = 1, // excluding clock and reset pads
    parameter int unsigned NUM_OUTPUT_PADS = 17,
    parameter int unsigned NUM_BIDIR_PADS  = 4,
    parameter int unsigned NUM_ANALOG_PADS = 4
) ( ... );
```

---

---

**Listing 4** rtl/chip\_top.sv (code snippet)

---

```
generate
    for (genvar i = 0; i < NUM_INPUT_PADS; i++) begin : g_inputs
        sg13g2_IOPadIn input_pad (
            `ifdef USE_POWER_PINS
                .iovdd (IOVDD),
                .iovss (IOVSS),
                .vdd    (VDD),
                .vss    (VSS),
            `endif
            .p2c    (input_PAD2CORE[i]),
            .pad     (input_PAD[i])
        );
    end
endgenerate
```

---

---

**Listing 6** rtl/chip\_core.sv (code snippet)

---

```
// =====  
// Input Assignments  
// =====  
logic enable;  
assign enable = input_in[0];  
  
// Inverter 1  
wire inv1_din1 = bidir_in[0];  
wire inv1_din2 = bidir_in[1];  
wire inv1_din3 = bidir_in[2];  
wire inv1_din4 = bidir_in[3];  
  
// Inverter 2  
wire inv2_vin1 = analog_padres[0];  
wire inv2_vin2 = analog_padres[1];  
// =====
```

---

---

**Listing 7** rtl/chip\_core.sv (code snippet)

---

```
// =====  
// Output Assignments  
// =====  
// Digital outputs  
// Counter 1  
assign output_out[0] = counter1_value[0];  
assign output_out[1] = counter1_value[1];  
assign output_out[2] = counter1_value[2];  
assign output_out[3] = counter1_value[3];  
  
// Counter 2  
assign output_out[4] = counter2_value[0];  
assign output_out[5] = counter2_value[1];  
assign output_out[6] = counter2_value[2];  
assign output_out[7] = counter2_value[3];  
assign output_out[8] = counter2_value[4];  
assign output_out[9] = counter2_value[5];  
assign output_out[10] = counter2_value[6];  
assign output_out[11] = counter2_value[7];  
  
// Inverter 1  
assign output_out[12] = inv1_dout1;  
assign output_out[13] = inv1_dout2;  
assign output_out[14] = inv1_dout3;  
assign output_out[15] = inv1_dout4;  
  
// SRAM  
// Reduce the 32-bit SRAM output to a single bit by computing the parity.  
// output_out[16] = 1 when an odd number of bits in sram_0_out are 1.  
// output_out[16] = 0 when an even number of bits in sram_0_out are 1.  
assign output_out[16] = ^sram_0_out;  
  
// Digital bidirectionals (output side)  
// Bidir output enable: drive when `enable` is high (counter visible),  
// float as input when low (so external stimuli can drive inverter1).  
assign bidir_out[0] = counter1_value[4];  
assign bidir_oe[0] = enable;  
  
assign bidir_out[1] = counter1_value[5];  
assign bidir_oe[1] = enable;  
  
assign bidir_out[2] = counter1_value[6];  
assign bidir_oe[2] = enable;  
  
assign bidir_out[3] = counter1_value[7];  
assign bidir_oe[3] = enable;  
  
// Analog outputs  
assign analog_padbare[2] = inv2_vout1;  
assign analog_padbare[3] = inv2_vout2;  
// =====
```

---

**Listing 8** flow/librelane/pdn\_cfg.tcl (code snippet)

---

```
# Custom connect for the inverter macros (inverter1, inverter2).
# The top-level power ring is on TM1 (horizontal) and TM2 (vertical).
# Each inverter has its own power ring on M5 (horizontal) and TM1 (vertical),
# so the connection between the top-level power ring and the inverter power rings crosses cl
# Note: within an inverter's power ring no top-level TM1 or TM2 power rails are routed.
define_pdn_grid \
    -macro \
    -instances "i_chip_core.inverter1 i_chip_core.inverter2" \
    -name inverter_top \
    -starts_with POWER \
    -halo "$::env(PDN_HORIZONTAL_HALO) $::env(PDN_VERTICAL_HALO)"

add_pdn_connect \
    -grid inverter_top \
    -layers "$::env(PDN_VERTICAL_LAYER) $::env(PDN_HORIZONTAL_LAYER)"

add_pdn_connect \
    -grid inverter_top \
    -layers "$::env(PDN_VERTICAL_LAYER) Metal5"
```

---